

Chapter 149 — Benign Prostatic Hyperplasia: Aetiology, Pathophysiology, Epidemiology and Natural History

- **Benign prostatic hyperplasia (BPH) is a histopathological process**, and it is one cause, but not the only cause, of lower urinary tract symptoms (LUTS) in ageing individuals, also termed male LUTS.
- **Cause and effect remain unestablished** after five decades of research. Androgens, for example, are **necessary but not sufficient** to drive BPH.
- **The old term "prostatism" is erroneous**. Male LUTS are not simply the result of a mass-related rise in urethral resistance.
- **A significant share of male LUTS is not prostatic**. Causes include age-related detrusor dysfunction, polyuria, sleep disorders and systemic disease unrelated to the prostate–bladder unit.
- **Voiding symptoms correlate poorly with the underlying pathophysiology**.
 - The same symptoms arise from **any outlet obstruction**, for example a urethral stricture.
 - They also arise from **impaired detrusor contractility**, with no obstruction at all.
 - LUTS are therefore often, but **not invariably**, due to bladder outlet obstruction (BOO) from benign prostatic obstruction (BPO). BPO in turn often accompanies benign prostatic enlargement (BPE) resulting from histological BPH.
- **Women have voiding symptoms too**. Age-matched community-dwelling men and women report **similar symptom frequency and severity**.
- **Failure to empty** results from outlet obstruction, detrusor underactivity, or both.
- **Post-micturition symptoms** (for example post-void dribbling) occur in both sexes. They are **commonest in men**, highly troublesome, and interfere significantly with quality of life (QoL).
- **Storage symptoms fall largely under overactive bladder (OAB) syndrome**.
 - OAB comprises urgency, frequency, nocturia and urgency incontinence, and is thought to reflect **detrusor overactivity**.
 - Storage symptoms are **more bothersome than voiding symptoms**, especially when accompanied by incontinence.
 - In both sexes they are commonly associated with **urinary infection**. Rarer associations are bladder stones, carcinoma and carcinoma in situ.
- **Four partially overlapping populations** (Figure 149.1): histological BPH, LUTS with bother, BPE (enlargement) and BOO (obstruction). The circles in the figure are not to scale.
 - **Many men over 40 develop histological BPH, but not all have bothersome LUTS**. Of those who do, some develop measurable enlargement (BPE) and some do not.
 - **BPE without LUTS, and LUTS without BPE, are both common**.
 - **BOO may exist with or without LUTS and with or without BPE**, and may occur in men **without BPH**, for example from urethral stricture.
- **Linking cell type-specific alterations to dysfunction** is the route to new strategies to treat, or prevent, the effect of BPH on lower urinary tract (LUT) function.

Concept — *BPH, BPE, BOO and LUTS are four different things (a histological process, an enlargement, an obstruction and a symptom complex). They overlap only partially, and nearly every argument later in the chapter depends on keeping them apart.*

AETIOLOGY OF BENIGN PROSTATIC HYPERPLASIA

- **The histology is hyperplasia, not hypertrophy.** BPH is an increased number of epithelial and stromal cells in the **periurethral** prostate. "Hypertrophy" in the older literature is incorrect.
- **New gland formation otherwise occurs only in fetal development.** This is the basis of the concept of **embryonic reawakening** of the inductive potential of the stroma.
- **Increased cell number may come from two sources:** proliferation, or impaired programmed cell death with cellular accumulation.
- **Candidate contributors**, acting singly or in combination: androgens, oestrogens, stromal–epithelial interactions, growth factors and neurotransmitters.

Hyperplasia

- **Organ volume reflects the balance between cell proliferation and cell death.** An organ enlarges if proliferation rises **or** if cell death falls.
- **Active proliferation is doubtful in established human BPH.** Early BPH may involve rapid proliferation, but established disease is maintained with an **equal or reduced replication rate**. Increased expression of **anti-apoptotic genes** (for example *bcl-2*) supports this.
- **Androgens have two actions:** they are required for normal proliferation and differentiation, and they **actively inhibit cell death**.
- **In the dog, androgen plus oestradiol produces experimental BPH.** Gland size rises markedly while **DNA synthesis actually falls**, showing that androgens and oestrogens both **inhibit cell death**.
- **Neural signalling may also balance death against proliferation**, especially α -adrenergic pathways.
- **Remodelling of architecture occurs in two stages.**
 - Early: **epithelial budding** from pre-existing ducts, and **mesenchymal nodules**.
 - Established disease: a **highly variable** tissue phenotype. This heterogeneity has confounded a cell type-specific molecular understanding.

Cellular identity and the normal prostate atlas.

- **BPH is highly pleomorphic.** Bulk tissue molecular analyses therefore track the patient-specific phenotype. **Single-cell RNA sequencing atlases** of normal human and mouse prostate now provide the baseline.
- **Architecture: a branching tree of 13–15 ducts on each side** of the prostatic urethra.
- **The prostatic urethral urothelium** contains **club cells (SCGB1A1+)** and **hillock cells (KRT13+)**, with interdigitating neuroendocrine cells and ionocytes. Club and hillock cells extend into the proximal ducts, then give way to **NKX3.1+ prostate luminal cells**. The same urethral-to-prostatic transition is seen in the mouse proximal ducts (Figure 149.2: keratin 5 marks basal epithelium, keratin 13 urethral hillock epithelium, NKX3-1 secretory luminal epithelium).
- **Glandular acini** are the functional units: secretory luminal and basal epithelium, with rare ionocytes and neuroendocrine, club and hillock cells.
- **Fibroblasts come in two populations.**
 - **Periepithelial fibroblasts** surround each acinus and also line the prostatic urethral urothelium.
 - **Interstitial fibroblasts** form a **dense collar around the prostatic urethra** from apex to base, and are sparse between peripheral zone glands.
- **Peripheral zone stroma is mostly smooth muscle.** It contracts the acini at ejaculation, driving fluid through the proximal ducts into the verumontanum.

- **Interstitial fibroblasts are increased in BPH**, in both stromal and glandular phenotypes. This effectively **expands the transition zone** and may induce new branching morphogenesis.
 - Their trigger is **unknown**.
 - They are **not targeted by any current medical therapy**.

BPH as a stem cell disease; regeneration after castration.

- **BPH recapitulates developmental branching morphogenesis** (budding → branching → branch elongation). This prompts the view of BPH as a **stem cell disease**, but the cells that form new glands are unidentified.
- **Mouse development**: multipotent basal progenitors are replaced by **lineage-committed unipotent basal and luminal progenitors**. The proximal (urethral) compartment derives from Nkx3.1-negative urethral epithelium marked by Krt4 and Runx1.
- **Both basal and luminal lineages can regrow the adult mouse prostate** after stress: inflammation, obesity, luminal anoikis or castration.
- **Castration shrinks the distal prostate** through loss of androgen-dependent secretory luminal cells. Androgen replacement regrows it **only to its original size**, implying a growth-restricting mechanism.
- **Two older hypotheses located the regenerating cell**:
 - An androgen-independent progenitor in the **distal** prostate, based on where DNA synthesis is concentrated.
 - Regenerative capacity in the **urethra and proximal ducts**, based on explant assays.
- **Modern lineage tracing resolves this**.
 - Urethral luminal cells (Sca-1+/Psc+/Trop2+/Krt4+) **survive castration but do not regenerate** the prostate.
 - **Most regeneration comes from surviving distal Nkx3.1+ luminal cells**. These survive low androgen by **lineage plasticity**, coming to resemble urethral luminal cells.
 - A rare pre-existing distal progenitor population contributes only a very small fraction.
 - Conclusion: **any surviving luminal cell can regenerate** the gland; there is no dedicated progenitor niche.
- **In men treated with a 5 α -reductase inhibitor (5ARI) or with androgen deprivation therapy**, prostate luminal cells undergo lineage plasticity to **resemble urethral club cells**.
- **In humans the origin of regrowth after androgen withdrawal is unresolved**. Three candidates:
 - Pre-existing 5ARI-indifferent SCGB3A1+ club cells in the proximal ducts.
 - 5ARI-induced club-like cells arising from luminal plasticity.
 - 5ARI-resistant NKX3.1+ cells.

Concept — *BPH is an accumulation disease. Cells gather because death is suppressed at least as much as division is driven, and androgens contribute mainly by keeping cells alive. That is why the gland shrinks when androgen is withdrawn.*

The Role of Androgens

- **Androgens are permissive, not causal**. BPH needs **testicular androgens** during development, puberty and ageing.
 - Men castrated **before puberty**, or with genetic defects of androgen production or action, **never develop BPH**.
 - **Androgen withdrawal causes partial involution** of established BPH.

- **Intraprostatic dihydrotestosterone (DHT) and androgen receptor (AR) levels stay high with age**, even though circulating testosterone falls.
- **Circulating androgens do not predict prostate size** within the normal range.
 - **Olmsted County cohort**: bioavailable testosterone falls with age, and the oestradiol:bioavailable testosterone ratio rises. Both associations with volume weaken markedly **after age adjustment**.
 - A large BPH medical-therapy study confirmed **no relationship among serum testosterone, prostate-specific antigen (PSA) and volume** (Table 149.1).
 - Exception: **higher baseline serum DHT** was associated with a higher risk of BPH over 20 years, and a **higher testosterone:DHT ratio** was associated with a **lower** risk.

TABLE 149.1 ABSENCE OF SIGNIFICANT RELATIONSHIP AMONG SERUM TESTOSTERONE AND SERUM PROSTATE-SPECIFIC ANTIGEN AND PROSTATE VOLUME (Modified from Marberger et al., 2006)

Baseline testosterone band	PSA (ng/mL)	Prostate volume (mL)	Body mass index (kg/m ²)	Sexual Function Inventory score	Age (years)
≥300 ng/dL (most men)	~4.0	~54	~27	~6.8	~66
175–300 ng/dL	3.8–4.0	55–57	28–30	5.9–7.2	65–67
<175 ng/dL	3.6–4.0	56–61	29.5–31	5.6–5.8	66–69

- **What the table shows**: across eight testosterone bands, PSA and volume barely move. Age is flat.
- **The one variable that tracks testosterone is body mass index (BMI)**, which rises as testosterone falls. Obesity lowers testosterone; testosterone does not set prostate size.
- **Unit trap**: testosterone in ng/dL; divide by 28.8 for nmol/L.

Mechanism of androgen action.

- **In brain, skeletal muscle and seminiferous epithelium, testosterone acts directly.**
- **In the prostate, nuclear membrane-bound 5α-reductase converts testosterone to DHT**, the principal prostatic androgen.
 - **About 90% of prostatic androgen is DHT**, derived mainly from testicular androgen.
 - **About 10% is adrenal**, and its role in BPH is negligible.
- **Testosterone and DHT bind the same high-affinity AR**. DHT is more potent because of **higher AR affinity**, and the DHT–receptor complex may be more stable.
- **Pathway**: hormone–receptor complex → DNA binding sites → transcription of androgen-dependent genes → protein synthesis.
- **Androgen withdrawal does three things**: switches off androgen-dependent genes (for example PSA), activates **programmed cell death** genes, and activates **inflammatory** genes.
- **Neither testosterone nor DHT is a direct mitogen.**
 - Neither is mitogenic to cultured prostatic epithelium.
 - Expression studies in the rat ventral prostate show no direct activation of mitogenic pathways.
 - Androgens act **indirectly, through autocrine and paracrine** growth factors and cytokines.
- **Figure 149.3 — the paracrine model.**
 - Testosterone enters stromal and epithelial cells.
 - In the **stromal cell** most is converted to DHT, which acts **autocrinely** or diffuses to adjacent epithelium (**paracrine**).
 - DHT made peripherally (**skin, liver**) reaches the prostate in a true **endocrine** fashion.
 - Basal cells may also produce DHT.

Androgen Receptors

- **The prostate keeps its androgen responsiveness for life; the penis does not.**
 - Penile AR falls to negligible levels at the **end of puberty**, so the adult penis stops growing despite high androgens.
 - **Prostatic AR remains high throughout ageing.** Nuclear AR may even be **higher in hyperplastic tissue**.
- **Age-related rises in oestrogen may up-regulate AR**, sustaining growth or reducing cell death despite falling serum androgen and "normal" intraprostatic DHT.
- **AR gene polymorphism is at most a minor factor.** Short CAG (cytosine–adenine–guanine) repeats were linked to larger glands and more surgery in one population, but findings are **inconsistent across populations**. Vitamin D receptor polymorphism findings are likewise inconsistent.

Dihydrotestosterone and Steroid 5 α -Reductase

- **Intraprostatic DHT is maintained, not elevated, in BPH.**
 - Early reports of raised DHT used **cadaveric (accident victim) controls**, in whom DHT falls after death.
 - Using **surgical specimens from men without BPH as controls**, DHT is the **same** in hyperplastic and normal glands.
 - The ageing prostate keeps **high DHT and high AR**, so the machinery for androgen-dependent growth persists. Androgens are at least **permissive**.

COMPARISON: THE TWO 5 α -REDUCTASE ISOENZYMES

Feature	Type 1	Type 2
Predominant site	Extraprostatic: skin and liver	Prostate (also some extraprostatic tissue)
Cell of expression in prostate	Barely detectable in BPH	Stromal cells, specifically interstitial fibroblasts ; some basal cells; luminal epithelium lacks it
5 α -reductase deficiency syndrome	Normally expressed	Mutated — this is the defective enzyme
Finasteride	Not substantially inhibited	Exquisitely sensitive
Dutasteride	Inhibited	Inhibited
Role	Possibly more relevant to prostate cancer	Critical for prostate development and for hyperplastic growth

- **Type 1–derived DHT is unlikely to matter in BPH.** Finasteride (type 2 only) shrinks the gland as much as dual inhibitors do, and roughly as much as castration.
- **Any extra benefit of dual inhibition** would come from blocking **peripherally produced DHT**.
- **The interstitial fibroblast is the androgenic amplification step.** Type 2 5 α -reductase within fibroblasts converts testosterone to DHT, which is the core of the **paracrine model**.
- **Enzyme expression varies within and between prostates**, so single biopsies map its distribution poorly.

Resistance to 5 α -reductase inhibitors.

- **5ARIs are the only medical therapy that reduces prostate volume**, which makes resistance an important question.
- **Clinical resistance is not failure to lower intraprostatic DHT.** More likely causes are:
 - **absence of luminal epithelial atrophy** focally within glandular nodules, leaving persistent anatomical obstruction, or
 - **widespread stromal hyperplasia**.
- **Proposed luminal survival mechanisms**, all linked to inflammation:
 - A **club cell transition** in areas of 5ARI-induced atrophy, with high nuclear factor- κ B (NF- κ B) activation.

- NF- κ B is raised in BPH and can induce **androgen-independent AR variants**.
- **Epigenetic silencing of SRD5A2** in a subset of patients, switching survival to **oestrogenic** pathways.
- **Inflammation correlates with 5ARI resistance** in clinical trials. All these pathways remain correlative.
- **A proper definition of resistance** should have three components: volumetric (volume unchanged), histological (no glandular atrophy) and biochemical (DHT decreased).

SRD5A2 genetics.

- **SRD5A2 lies on chromosome 2p23**. It carries **A49T** and **V89L** substitutions and a thymine-adenine (TA) dinucleotide repeat.
 - The **89L allele lowers** enzyme activity; the **49T allele raises** it.
 - **Longer TA repeats** make the messenger RNA unstable and lower activity.
- **Links to BPH are inconsistent**. In the Olmsted County population there was no consistent genotype association, only a weak V89L-volume correlation.
- **Promoter methylation of SRD5A2** occurs in up to **30%** of BPH specimens, is thought to switch regulation to oestrogenic pathways, and **does not correlate with volume**.

Vascular effects.

- **Castration causes acute, drastic prostatic vasoconstriction** in the rat. This is **not mediated by vascular endothelial growth factor (VEGF)**.
- **Vascular disease elsewhere (for example diabetes)** may be a risk factor for BPH.

Concept — *Androgen levels do not rise in BPH. What persists is the machinery: high intraprostatic DHT, high AR, and type 2 5 α -reductase in fibroblasts. That is why DHT suppression shrinks the gland even though serum testosterone does not predict its size.*

The Role of Oestrogens

- **Oestrogen's role is clear in animals and uncertain in humans.**
- **In the dog, oestrogen synergises with androgen**. It **induces AR** and "sensitises" the ageing gland to androgen. The canine prostate is rich in high-affinity oestrogen receptor (ER).
- **Canine stromal effect**: oestrogen stimulates the stroma and increases total **collagen**.
- **The two receptors sit in different compartments: ER α in stroma, ER β in epithelium**. The prostate's response depends on which receptor is present.
- **Knockout mice** suggest oestrogens exert a "**constraining influence**" on the prostate.
- **In vitro**, stromal ER α up-regulation raises fibroblast growth factor (FGF)-2, FGF-7 and other growth factors. **Adding androgen down-regulates** both ER α and these factors.
- **ER β agonism failed clinically**. A randomised placebo-controlled trial in LUTS showed **no effect** on symptoms, PSA, volume or urodynamics.
- **Serum oestrogen rises with age**, absolutely or relative to testosterone.
 - Intraprostatic oestrogen may be raised in BPH.
 - Larger glands go with **higher serum oestradiol**.
 - **Olmsted County**: in men with above-median bioavailable testosterone, oestradiol correlated positively with volume **even after age adjustment**.
 - Data linking obesity, testosterone, oestradiol and volume **conflict**.
- **Classic high-affinity ER is sparse in human BPH**, but possibly sufficient for activity.
- **Aromatase inhibition** reduces drug-induced stromal hyperplasia in animals.
- **Species differences and uncertain cause and effect** keep the human role unproven.

TABLE 149.2 COMPARISON OF ESTROGEN RECEPTOR-A AND ESTROGEN RECEPTOR-B EXPRESSION AND ACTIVITIES IN THE PROSTATE GLAND (From Prins and Korach, 2008)

Property	ER α	ER β
Localisation	Stromal	Epithelial
Proliferation	Stromal proliferation; epithelial squamous metaplasia	Antiproliferative
Differentiation	Epithelial dysplasia	Prodifferentiation
Immune response	—	Anti-inflammatory
Other	—	Antioxidant
Expression in prostate cancer	Dysregulated	Dysregulated: silenced in early cancer , reduced in organ-confined disease, re-emerges with progression , raised in metastatic disease

- **What the table is for:** the two receptors oppose each other. ER α is growth-promoting and stromal; ER β is restraining, differentiating and epithelial.
- **Therapeutic logic:** receptor-selective agonists and antagonists, for example ER β agonism. This has so far **failed** in LUTS.
- **Trap:** do not assign ER α to the epithelium. **α = stroma, β = epithelium.**
- **The progesterone receptor is abundant** in normal and hyperplastic prostate, but its role is **undefined**.

Concept — *Oestrogen's effect depends on which receptor carries it: stromal ER α promotes growth and epithelial ER β restrains it. Opposing receptors in adjacent compartments explain why animal data are strong and human data inconsistent.*

Regulation of Programmed Cell Death

- **Apoptosis maintains glandular homeostasis.**
 - Sequence: cellular condensation → fragmentation → phagocytosis by neighbouring cells → lysosomal degradation.
 - It occurs **without immune activation** but **requires RNA and protein synthesis**.
- **In the rat, cell death normally occurs in the proximal ductal segment** despite normal testosterone. Androgens suppress death **elsewhere** in the gland.
- **After castration, death rises** in the luminal epithelium and in the distal region of each duct.
- **Androgen action is regionally controlled** through local growth regulatory factors. The transforming growth factor- β (TGF- β) family is the likely mediator.
- **At least 25 genes** are induced in the rat prostate after castration.
- **Homeostasis balances growth inhibitors against mitogens.** A local growth factor or receptor abnormality could produce BPH through more proliferation **or less programmed death**.

Concept — *Androgens control prostate size largely by controlling death rather than division. Withdrawal therefore triggers an active, gene-dependent apoptotic programme rather than passive shrinkage.*

Stromal–Epithelial Interaction

- **Stroma and epithelium communicate in a paracrine fashion.**
- **Culture experiment in canine epithelium:**
 - Grown on **plastic**, the cells lose secretion, grow rapidly and change cytoskeletal staining.
 - Grown on **prostatic collagen**, they keep normal secretion and phenotype and grow slowly.

- So the **extracellular matrix (ECM)** partly regulates epithelial differentiation.
- **Hypothesis:** BPH may reflect a stromal defect that **removes a normal "brake"** on proliferation. The same defect could act autocrinely to expand the stroma.
- **Embryonic mesenchyme dictates differentiation of urogenital sinus epithelium.** New gland formation in BPH suggests **reawakening** of this stromal inductive process.
- **Growth factors and the ECM**, which has growth factor-like properties, mediate much of this. The model fits **5 α -reductase localisation in the periurethral interstitial fibroblast collar**.
- **CXC motif chemokines (CXCL1, CXCL5, CXCL6, CXCL12):**
 - They are made by the microenvironment and stimulate **both** stroma and epithelium.
 - With interleukin (IL)-8, they convert **fibroblasts to myofibroblasts**, which cause collagen deposition and **fibrosis**. Fibrosis may contribute to LUT dysfunction.
- **A putative stem cell niche** of poorly defined stromal and epithelial cells is maintained by **Hedgehog, Notch, WNT and TGF- β** signalling. These are potential targets for inducing regression.

Concept — *In the prostate the stroma instructs the epithelium, as embryonic mesenchyme does. BPH can therefore be read as the stroma resuming its fetal inductive role.*

Growth Factors

- **Growth factors are small peptides** that stimulate or inhibit division and differentiation through surface receptors. Their interaction with steroid hormones may tip proliferation against death (Figure 149.4).
- **The first prostatic growth factor was basic fibroblast growth factor (bFGF, FGF-2)**, identified by sequencing an extract of BPH that stimulated growth.
- **Factors implicated:** FGF-1 (acidic), FGF-2 (basic), FGF-3 (Int-2), **FGF-7 (keratinocyte growth factor, KGF)**, FGF-10, FGF-17, TGF- β , epidermal growth factor (EGF), VEGF and insulin-like growth factors (IGFs).
- **Figure 149.4 message:** androgens are **necessary but permissive**; growth factors are the likelier site of the **primary defect**.
- **The overall role is still uncertain.** Growth factors probably matter, but proliferation is not obvious in BPH, and it is unclear which factors are key mediators and which are bystanders.

COMPARISON: GROWTH FACTORS IN BPH

Factor	Source	Target / action	BPH evidence
FGF-2 (bFGF)	Stroma	Autocrine stromal mitogen	Raised 2–3-fold in BPH; transgenic overexpression → epithelial hyperplasia; induced by TGF- β and hypoxia
FGF-7 (KGF)	Stroma (adjacent to epithelium)	Receptor exclusively on epithelium → paracrine	Overexpressed in BPH; transgenic overexpression → atypical hyperplasia; leading candidate stromal mediator of hormonal control of epithelium ; induced by epithelial IL-1α
FGF-10	Stromal smooth muscle	Epithelial mitogen	Increased by androgens
FGF-17, VEGF	—	Mitogenic	Implicated
TGF-β1	Stroma	Inhibits epithelium; mitogen for fibroblasts ; regulates ECM; induces apoptosis; up-regulates FGF-2; autocrine mitogen for a prostate smooth muscle line	BPH stroma shows blunted TGF-β growth inhibition (reduced IGFBP-3 induction); up-regulation → stromal expansion ; codon 10 polymorphism → higher BPH risk

Factor	Source	Target / action	BPH evidence
IGF-1, IGF-2	—	Prostate growth in culture	Transgenic IGF-1 → enlargement; IGF-2 higher periurethrally; serum IGF-1 not related to volume in Olmsted County

- **TGF- β has opposite actions by compartment:** it restrains epithelium and drives stroma. Loss of epithelial restraint plus stromal stimulation fits a **stroma-dominant** BPH.
- **TGF- β also regulates** smooth muscle contractile proteins, and causes **versican** accumulation in the ECM by inhibiting the ADAMTS metalloproteases that degrade it.
- **FGF-7 is the paracrine answer** to how an androgen signal in the stroma reaches the epithelium: stromal source, epithelial receptor.
- **Hypoxia hypothesis:**
 - Local hypoxia (for example from atherosclerosis) → **hypoxia-inducible factor 1** → stromal secretion of FGF-2 and FGF-7 → nodule growth → further impaired blood flow → more hypoxia.
 - TGF- β , itself activated by hypoxia, also regulates FGF-2.
- **Trap:** TGF- β is not simply "growth inhibitory". It inhibits epithelium but is **mitogenic for fibroblasts**.

Concept — Androgen supplies the permission and growth factors supply the signal. The factors are compartment-specific: FGF-7 carries a stromal message to the epithelium, and TGF- β brakes the epithelium while driving the stroma.

Other Signalling Pathways

- **Sympathetic signalling matters in LUTS.** α -Adrenergic blockers are highly effective for LUTS.
- **Sympathetic pathways may also drive growth.** α -Blockade can **induce apoptosis** in some models, and α -adrenergic pathways modulate the smooth muscle phenotype.
- **The renin-angiotensin system (RAS):** all its components are present in the prostate and may be activated in BPH. Local RAS may contribute to proliferation and smooth muscle contraction, with or without sympathetic modulation.
- **Early growth response gene 1 (EGR-1)** is active in a BPH cell line.
- **α 2-Macroglobulin** binds PSA and many growth factors, is highly expressed in the prostate, and is **up-regulated in BPH**. Trapping inhibitory molecules could promote growth.

The Potential Role of Inflammatory Pathways and Cytokines in Benign Prostatic Hyperplasia

- **Inflammatory infiltrates are an additional source of growth factors.**
 - BPH tissue shows extensive infiltration by **activated T cells**.
 - T cells secrete VEGF (a potent epithelial mitogen), heparin-binding EGF and FGF-2, and so could drive stromal and glandular hyperplasia.
- **Cytokines in BPH tissue:**
 - **IL-2, IL-4, IL-7, IL-17 and interferon- γ** and their receptors are present. **IL-2, IL-7 and interferon- γ stimulate stromal proliferation** in vitro.
 - **Senescent epithelium secretes IL-8**, which promotes proliferation of non-senescent epithelial and stromal cells.
 - **Macrophage inhibitory cytokine 1** is present in normal prostate and **down-regulated in BPH**.
 - Chronic inflammation is associated with focal epithelial **cyclo-oxygenase-2 (COX-2)** up-regulation.

- **Mouse models support a link.**
 - The **POET (prostate ovalbumin-expressing transgenic) autoimmune model** shows lymphocyte, macrophage and neutrophil infiltration with epithelial and stromal thickening and **epithelial stem cell expansion**.
 - Bacterial prostatitis drives epithelial turnover.
 - In the human prostate, epithelial **progenitors expand near inflammation**.
 - Even so, **no firm cause and effect** has been established.
- **The metabolic syndrome may raise the risk of prostatic inflammation**, and so of LUTS and BPH.
- **Androgens are anti-inflammatory.**
 - AR activation by testosterone (rabbit) and DHT is protective.
 - Conditional AR knockout in mouse prostate epithelium → more proinflammatory cytokines and immune infiltration.
 - Human **hypogonadism is associated with inflammation**.
 - If confirmed, this links androgen status to inflammation in male LUTS and BPH.
- **BPH as an autoimmune or immune-inflammatory disease** (Figure 149.5):
 - T cell–derived proinflammatory cytokines drive immune inflammation and stromal growth.
 - **Smooth muscle cells help maintain and propagate** the infiltrate.
 - Antigens may be prostatic (PSA, prostatic acid phosphatase) or infectious (bacteria, viruses, protists, fungi).
 - Inflammation has been proposed as a **therapeutic target**.
- **Single-cell sequencing: fibroblasts are the central communicators** with the immune infiltrate. The influx of T cells, B cells and dendritic cells resembles a **mucosal inflammatory disorder**.
- **Autoimmune link in a large database:**
 - BPH prevalence was **higher in patients with autoimmune disease**.
 - **Tumour necrosis factor (TNF) antagonist therapy was associated with lower BPH incidence**.

Concept — *Androgens and inflammation pull in opposite directions: androgen is anti-inflammatory and hypogonadism is pro-inflammatory. Inflammation, not androgen excess, is the more plausible accelerant of the ageing gland, but causality remains unproven.*

Genetic and Familial Factors

- **BPH has a heritable component.**
- **Johns Hopkins case-control study:**
 - Cases were men undergoing surgery with **resected weight in the top quartile (>37 g)** and **age at surgery in the bottom quartile**.
 - First-degree male relatives of cases had about **fourfold** the risk of surgically treated BPH, not explained by health-seeking behaviour (Table 149.3).
 - **Segregation analysis favoured autosomal dominant inheritance**.
 - About **50% of men operated on before 60** may have heritable disease, against about **9% of those operated on after 60**.
- **Twin data: monozygotic concordance exceeds dizygotic.**
- **Community cohort:** a family history of enlarged prostate or BPH raises the risk of moderate-to-severe symptoms.
- **Finasteride trial families (three or more affected members):**
 - Familial BPH is characterised by **substantially larger glands** than sporadic BPH.

- **Serum androgens and 5ARI response are the same** in familial and sporadic disease.
- Inheritance tracks **early onset and large volume** more than symptom severity.
- **UK Biobank:** genetic risk of **BPH and prostate cancer are highly correlated** (risk single-nucleotide polymorphisms).
- **Other molecular findings:** DNA mutations and hypomethylation, abnormal nuclear matrix proteins, assorted polymorphisms and **abnormal Wilms tumour gene (WT-1)** expression. Prostate cancer variants at 2q31 and 5p15 have been associated with **aggressive BPH** in a Chinese population.
- **Risk genes probably act in combination;** no single causal mutation has been identified.

TABLE 149.3 FAMILY HISTORY OF EARLY-ONSET BENIGN PROSTATIC HYPERPLASIA INCREASES RISK OF CLINICALLY SIGNIFICANT BENIGN PROSTATIC HYPERPLASIA (From Sanda et al., 1994)

Relatives of proband	Clinical BPH: case vs control relatives	Unadjusted odds ratio (95% CI)	Relative risk, Cox model (95% CI)	Significance
All first-degree male relatives	~28% vs ~9%	4.2 (1.7–10.2)	4.4 (1.9–9.9)	Strong
Fathers	~33% vs ~13%	3.3	3.5	Significant
Brothers	~24% vs ~4%	8.0	6.1	Significant

- **Outcome definition:** prostatectomy (open or transurethral) for BPH among informative male relatives.
- **Pattern:** the risk is **highest in brothers**, consistent with a shared early-onset inherited trait.
- **Two measures, two values:** the text describes the all-relatives figure as a hazard ratio of 4.2, which is the table's **unadjusted odds ratio**. The **Cox proportional hazards** relative risk is **4.4**. The difference is the statistical model, not the data.
- **Trap:** familial BPH is defined by **young age at surgery and large gland**, not by severe symptoms. It shows **normal androgen levels and normal 5ARI response**.

Concept — *Heritable BPH is early-onset, large-volume disease with dominant inheritance. The younger the man coming to surgery, the more likely his disease is familial.*

Other Aetiological Factors

- **Only dogs and humans develop BPH**, although all mammalian prostates have testosterone, DHT, AR and the usual growth factor pathways. The **seminal vesicle**, also androgen-responsive for life, never becomes hyperplastic. **Species-specific cofactors** must exist.
- **Testicular non-androgenic factors:**
 - They may reach the prostate through the vas deferens or deferential vessels.
 - **Testis-intact** rats and dogs given androgens grow more prostate than castrated ones given the same androgen, and intact dogs show more histological BPH.
 - **Human spermatocele fluid is mitogenic** to prostatic epithelium and stroma.
- **Prolactin:** transgenic overexpression enlarges the mouse prostate. Prolactin receptors are present in man, but the human role is **unclear**.
- **No mouse model of human BPH exists.** Mouse and human anatomy differ, and the molecular drivers are poorly understood. Models of specific aspects are needed.

Concept — *The androgen and growth factor machinery is universal across mammals, yet only men and dogs develop BPH. Something species-specific, possibly testicular, must be added to that universal machinery.*

PATHOPHYSIOLOGY

- **The pathophysiology is a chain, not a single lesion** (Figure 149.6): prostatic hyperplasia → raised urethral resistance → compensatory change in bladder function.
- **The price of compensation is storage function.** The higher detrusor pressure needed to maintain flow against resistance is achieved at the expense of normal storage.
- **Obstruction-induced detrusor change, compounded by ageing of bladder and nervous system,** produces frequency, urgency and nocturia, the most bothersome BPH-related complaints.
- **Figure 149.6 — the interacting inputs to LUTS:** BOO (from hyperplasia or from non-BPH causes), the detrusor response to obstruction, detrusor ageing effects, neurogenic disease, primary bladder disease and polyuria.
- **Understanding BPH therefore requires understanding obstruction-induced bladder dysfunction.**

Pathology

Anatomic Features

- **BPH begins in the periurethral transition zone** (McNeal's zonal anatomy, Figure 149.7).
- **The transition zone** is two separate glands lying immediately **outside the preprostatic sphincter**. Its main ducts arise on the **lateral aspects of the urethral wall at the point of urethral angulation near the verumontanum**.
- **The periurethral zone** lies **proximal** to the transition zone duct origins, **confined within the preprostatic sphincter**, with glands running **parallel to the urethral axis**.
- **Every BPH nodule arises in either the transition zone or the periurethral region.**
 - Early transition zone nodules lie within or immediately beside the preprostatic sphincter.
 - As nodules multiply they appear almost anywhere in the transition or periurethral zone.
 - **The transition zone also enlarges with age independently of nodule formation.**
- **Figure 149.7** shows the zones in sagittal and coronal section: peripheral, transition and central zones, verumontanum, proximal urethral segment, preprostatic sphincter and ejaculatory duct.

The prostatic capsule.

- **The capsule is unique to the human prostate and is central to LUTS.**
 - **The dog** develops natural BPH but **rarely develops obstruction or urinary symptoms**, because its prostate **lacks a capsule**.
 - The capsule presumably **transmits the pressure of tissue expansion to the urethra**, raising urethral resistance.
 - **Clinical proof: transurethral incision of the prostate (TUIP)** incises the capsule and **relieves obstruction significantly although prostate volume is unchanged**.
- **Prostate size does not correlate with the degree of obstruction.** Dynamic urethral resistance, the capsule and anatomical pleomorphism matter more than absolute gland size.

Lobes (Figure 149.8).

- **Hyperplastic tissue obstructs the urethra in recognisable patterns:** isolated **middle lobe**; isolated **lateral lobes**; **lateral plus middle lobes**; and **posterior commissural hyperplasia (median bar)**.
- **The middle lobe must be periurethral in origin**, because **there is no transition zone tissue at the bladder neck**. It results from predominant growth of periurethral nodules at the bladder neck.
- **Whether middle lobe growth is random or genetically determined is unknown.**

Histologic Features

- **BPH is hyperplasia:** a net increase in **cell number**, not cell size. Histology shows increased cell numbers, and thymidine uptake in canine experimental BPH shows **increased DNA synthesis**. "Benign prostatic hypertrophy" is **pathologically incorrect**.

COMPARISON: THE TWO EARLY NODULE TYPES

Feature	Periurethral (stromal) nodules	Transition zone (glandular) nodules
Composition	Purely stromal in most early nodules	Glandular proliferation , with a relative reduction in stroma
Appearance	Resemble embryonic mesenchyme : abundant pale ground substance, minimal collagen	Tightly packed glands, taller lining epithelium, some epithelial cell hypertrophy
Cell of origin	Interstitial fibroblasts predominate (single-cell RNA sequencing)	New small duct branches budding from existing ducts , forming an entirely new ductal system within the nodule
Stroma present	—	Scant, and mostly mature smooth muscle , like uninvolved transition zone

- **New gland formation of this kind is otherwise rare outside embryonic development.**
- **Transition zone volume rises with age** through more nodules and through enlargement of the whole zone.

Two phases of nodular evolution.

- **Phase 1 — the first 20 years:** mainly an **increase in nodule number**; each nodule grows slowly. Glandular nodules are already larger than stromal ones.
- **Phase 2:** a marked increase in **large nodules**. Growth of individual nodules now dominates, and **glandular nodules clearly predominate in size**.

Stromal–epithelial ratio.

- **Stromal-to-epithelial ratios vary widely** between resected specimens, a pleomorphism largely ignored in molecular studies.
 - **Small resected glands** show **predominantly fibromuscular stroma**.
 - **Large glands, mostly removed by enucleation**, show **predominantly epithelial nodules**.
- **A high stromal ratio does not prove a "stromal disease":** stromal proliferation may itself be driven by epithelial disease.
- **Figure 149.9** shows stromoglandular hyperplasia, with glandular and stromal tissue side by side.

Importance of Prostatic Smooth Muscle

- **Smooth muscle is a major fraction of gland volume.** In symptomatic BPH, the area density of **smooth muscle and connective tissue significantly exceeds that of glandular epithelium and lumen** (Figure 149.10).
- **Its spatial arrangement is not optimal for force generation**, yet passive and active forces both play a major part in BPH.

COMPARISON: THE TWO COMPONENTS OF PROSTATIC URETHRAL RESISTANCE

Feature	Active (dynamic) tone	Passive tone
Source	Adrenergic stimulation of smooth muscle	Series elastic elements of stromal and epithelial cells and, most importantly, the extracellular matrix

Feature	Active (dynamic) tone	Passive tone
Receptor	α1a-adrenoceptor : the most abundant subtype, and the mediator of active tension	Independent of active contraction
Effect of α -blockade	Diminished	Not reduced
Status	Well characterised	Determinants still unknown; may be an equal determinant of resistance

Stroma and androgen withdrawal.

- **Stroma is usually assumed to resist androgen withdrawal**, and short-term ablation affects mainly the epithelium.
- **Stromal turnover is much slower than epithelial turnover.** If ablation acts by raising cell death, a fall in stromal cell number may not show **until a year or more** of therapy. True stromal resistance is therefore unproven.
- **Hormonal therapy may alter stromal function without reducing stromal volume.** Contractile proteins, neuroreceptors and ECM proteins are hormone- and growth factor-regulated in other smooth muscle systems, and **androgens modulate α -agonist effects** on prostatic smooth muscle in vitro.

Phenotype of BPH smooth muscle.

- **BPH smooth muscle is dedifferentiated.** Smooth muscle myosin heavy chain is **down-regulated** and non-muscle myosin heavy chain **up-regulated**. This indicates proliferation or loss of normal modulation.
- **α -Blockade in BPH further down-regulates normal contractile protein gene expression**, specifically smooth muscle myosin heavy chain. Adrenergic signalling therefore regulates the muscle cell, not merely its contraction.
 - Parallels: adrenergic transmitters regulate contractile gene expression and hypertrophy in cardiac muscle.
 - Testosterone may regulate adrenoceptor expression, at least in the kidney.

Other contractile regulators.

- **α 1-Adrenoceptor nomenclature has been standardised** to reconcile pharmacological and molecular classifications.
- **Endothelin and its receptors** are present in the prostate; their physiological role is undefined.
- **The kallikrein-kinin system (for example bradykinin)** may regulate smooth muscle proliferation and contraction.
- **Phosphodiesterase (PDE) type 4 and type 5 isoenzymes** are present in the prostate **and the detrusor**. **Placebo-controlled trials confirm that erectile dysfunction (ED) drugs benefit men with LUTS and BPH.**

Autonomic overactivity.

- **Autonomic nervous system activity correlates positively with symptom score** and other BPH measures, assessed by physiological tests and plasma and urinary catecholamines.
- **The rise in serum noradrenaline after tilt predicted transition zone size.**

Concept — *Obstruction in BPH depends on geometry and tone, not on mass. The capsule, lobe pattern and active and passive smooth muscle tone determine urethral resistance, which is why gland size and obstruction do not correlate.*

The Bladder's Response to Obstruction

- **The bladder's response is largely adaptive**, but many LUTS come from **obstruction-induced bladder change rather than from the obstruction itself**.

- **About one-third of patients keep significant voiding dysfunction, mostly storage symptoms, after surgical relief of obstruction.**

COMPARISON: THE TWO TYPES OF OBSTRUCTION-INDUCED BLADDER CHANGE

	Storage-type change	Emptying-type change
Change	Detrusor instability or reduced compliance	Reduced detrusor contractility
Symptoms	Frequency and urgency	Weaker stream, hesitancy, intermittency, rising residual urine
End stage	—	Detrusor failure (a minority)

- **Acute urinary retention (AUR) is not the inevitable end of this process.** Many men presenting with AUR have **more than adequate detrusor function** and an identifiable **precipitating event**.

Trabeculation and extracellular matrix.

- **Trabeculation, the main endoscopic change, is due to increased detrusor collagen.**
- **Severe trabeculation goes with significant residual urine**, suggesting incomplete emptying may reflect **collagen rather than impaired muscle**. Severe trabeculation is, however, a feature of **advanced** disease.
- **There is no universal definition or accepted grading system** for trabeculation; one recent grading attempt correlated acceptably with urodynamics.
- **Unrelieved obstruction increases detrusor ECM (collagen)** in animals, and probably in humans, though cause and effect are unproven.

The detrusor smooth muscle cell.

- **The first response to obstruction in animals is smooth muscle hypertrophy.** It is adaptive, maintaining flow against raised pressure, but brings intracellular and extracellular changes that lead to **instability** and, in some, **impaired contractility**.
- **Obstruction also causes** altered contractile protein expression, **mitochondrial dysfunction** (impaired energy production), calcium signalling abnormalities and impaired cell-to-cell communication.

COMPARISON: RESPONSE TO LOAD — SKELETAL VERSUS DETRUSOR MUSCLE

Skeletal muscle	Detrusor smooth muscle
A normal repertoire of contractile genes is up-regulated	Myosin heavy chain isoform switch and altered thin filament-associated proteins
More normally organised contractile units assemble	Reversion to a secretory phenotype
Adaptive	Less adaptive: the phenotypic switch increases ECM production

Neural and ageing effects.

- **Obstruction also modulates neural–detrusor responses.**
- **Ageing rats show altered neural control of micturition:** reduced contractility, impaired central processing and altered sensation.
- **Ageing alone reproduces some of the same changes** in bladder function, histology and cell function, independent of obstruction.
- **Ischaemia:** in animals, atherosclerosis and chronic bladder ischaemia or hypoxia (including from raised wall tension) may contribute to bladder pathology.

Three-stage model of bladder wall remodelling.

- **Hypertrophy → compensation → decompensation.**

- **Every layer takes part:** urothelium, suburothelium, detrusor smooth muscle, detrusor ECM and bladder nerves.
- **Stimuli:** cyclic stretch, raised hydrostatic and cyclic hydrodynamic pressure, and **hypoxia**.
- **A late result of long-standing obstruction is detrusor underactivity (underactive bladder):** an **end-stage condition** with limited therapeutic options, increasingly prevalent in the elderly.

Concept — *The detrusor adapts badly. Load drives a secretory, collagen-producing phenotype rather than normal hypertrophy, so obstruction leaves damage that surgery on the prostate cannot reverse. This explains the storage symptoms that persist after surgery.*

KEY POINTS — AETIOLOGY AND PATHOPHYSIOLOGY

- Androgens are ****necessary for maintenance but not growth**** of the prostate, and are ****anti-inflammatory****. - ****Inflammation correlates with prostatic enlargement****; its causal role is unclear. - ****Anatomical pleomorphism within and between patients**** confounds cell type-specific understanding of BPH and may contribute to ****therapeutic resistance****. - ****Animal models do not fully recapitulate human BPH****, because of anatomical differences and poor mechanistic understanding. - ****A heritable form of BPH exists****; the precise mutations are unidentified. - ****Understanding the secondary effects of obstruction on bladder contractile and neuromodulatory function**** could improve clinical outcomes in BPH and LUTS.

EPIDEMIOLOGY AND NATURAL HISTORY

Definitions

- **Epidemiology studies the distribution and determinants of disease.**

COMPARISON: THE TWO BRANCHES OF EPIDEMIOLOGY

Descriptive epidemiology	Analytical epidemiology
Describes incidence, mortality and prevalence by person, place and time	Searches for determinants of risk , with prevention in view
Compares rates within a population (by sex, age, socioeconomic group) and between populations (culture, ethnicity, lifestyle, diet)	Tests candidate risk factors

RATES

Rate	Definition
Incidence	New cases per 100,000 population per year
Prevalence	Existing cases per 100,000 population at a given date
Mortality	Deaths per 100,000 population per year
Fatality	Deaths per number of diseased

- **For a benign condition such as BPH, prevalence matters more than incidence.** Incidence is the key rate for highly fatal conditions.
- **There is no globally accepted epidemiological definition of BPH.** Every prevalence or incidence figure must be read against the investigator's definition.
 - **Histological (autopsy) prevalence** is easier to compare across studies.
 - **Clinical prevalence** depends on widely varying clinical definitions.
- **BPH mortality and fatality are now low**, but the historical data remain instructive.

COMPARISON: STUDY DESIGNS IN DESCRIPTIVE EPIDEMIOLOGY

Cross-sectional	Longitudinal
A population, stratified by baseline parameters, is assessed once	A population is assessed at baseline and at regular intervals
Cheaper and logistically easier; most LUTS and BPH studies	Costly and difficult to sustain
—	Longitudinal autopsy studies are impossible by definition ; histological longitudinal studies are exceedingly difficult because tissue must be obtained repeatedly

Descriptive Epidemiologic Studies**Histologic or Autopsy Prevalence**

- **Definition:** stromoglandular hyperplasia in a surgical specimen, or in whole prostates removed at autopsy from men who did not die of prostatic disease.
- **Landmark pooled autopsy study (Berry, 1984):** no BPH under 30; prevalence rises with each age group, peaking at 88% in men in their 80s.
- **Figure 149.11 — pooled autopsy series** (United States, England, Austria, Norway, Denmark, China, Japan, India): prevalence rises rapidly from the fourth decade and approaches 100% in the ninth decade.
- **Age-specific autopsy prevalence is strikingly similar across all ethnic and geographic populations.**

Cross-Sectional Studies of Clinical Prevalence

- **Descriptive epidemiology needs a single accepted disease definition, and BPH has none.**
- **"Prostatism" is obsolete and incorrect.** It wrongly made the prostate the sole source of LUTS. Paul Abrams coined "LUTS" to replace it.
- **Clinical classification uses three independent axes:**
 - **Symptoms:** mild, moderate or severe on a standardised questionnaire.
 - **Enlargement:** by digital rectal examination (DRE), transrectal ultrasonography (TRUS) or magnetic resonance imaging (MRI).
 - **Obstruction:** by flow rate or invasive pressure–flow study.
- **BPH is only one cause of LUTS.** Others are bladder, urethral, neurogenic and unknown causes.
- **Obstruction may exist without BPH:** urethral stricture, prostate cancer, primary bladder neck sclerosis (Figure 149.1).
- **Bother, interference with daily activities and impact on QoL** distinguish patients beyond symptom counts.
- **"Clinical BPH" is an imprecise term** for this constellation of LUTS, bother, interference and QoL impact, with or without enlargement or obstruction.
- **Few clear cut-offs separate disease states.** A volume over 30 mL is one proposed definition of clinical BPH, but higher and lower cut-offs are argued, and the same applies to symptom and obstruction thresholds.
- **What is described is therefore the age-stratified distribution of disease attributes**, not true disease prevalence.
- **Figure 149.12 — prevalence varies with the definition used:** autopsy, clinical diagnosis, low maximum flow rate, DRE enlargement and community-based composite definitions give different estimates. A common composite was **IPSS >7, Qmax <15 mL/s and prostate volume >20 mL.**

Symptom Severity and Frequency

- In a rarely fatal disease defined by its effect on QoL, symptom studies matter most.
- The American Urological Association (AUA) Symptom Index, also called the International Prostate Symptom Score (IPSS): a standardised, self-administered **seven-item** instrument, culturally and linguistically validated. Its introduction was **pivotal** for LUTS research.

IPSS SEVERITY BANDS

Band	Score
Mild	0–7
Moderate	8–19
Severe	20–35

- **Total range 0–35.** The IPSS is integral to virtually all epidemiological and treatment studies, and validated translations permit cross-cultural comparison.
- **Responses are robust:** unaffected by socioeconomic factors, and fundamentally the same whether self-administered, read aloud, mailed or re-administered in scrambled order.
- **Cross-cultural differences in reported severity** arise partly from comprehension of translations, symptom perception, willingness to admit symptoms and acceptance of symptoms as normal ageing.
- **Figure 149.13 — worldwide cross-sectional studies:** the proportion with at least moderate symptoms differs between countries, but **every study shows severity rising with age**. A large pan-Asian study gave results in line with European and North American data.
- **Main shortcoming of the IPSS: no items on incontinence or pain**, although stress and urge incontinence and post-void dribbling are among the commonest and most bothersome symptoms of ageing men.
- **The Lower Urinary Tract Research Network (LURN) questionnaire** (convened by the National Institute of Diabetes and Digestive and Kidney Diseases) addresses this.
 - Full **29-item** version (LURN SI-29) and abbreviated **10-item** version (LURN SI-10).
 - It covers **incontinence, bladder pain and post-micturition symptoms**, and applies to **both sexes**.
 - It correlates well with the IPSS and other questionnaires, but is **not yet widely used**.
 - **Figure 149.14:** LURN SI-29 domain scores (incontinence, pain, voiding difficulty, urgency, nocturia, total) rise with AUA symptom severity category in both men and women.

Bother, Interference, and Health-Related Quality of Life

- **Bother and interference are at least as important as symptom counts.**
- **Embarrassment is an important determinant of seeking care.**
- **General QoL measured by the Medical Outcomes Trust Short-Form (SF-36) is affected only marginally** by LUTS and clinical BPH. This prompted **disease-specific** instruments for bother, interference, QoL and sexual function.
- **These instruments are used less often than the IPSS** in cross-sectional studies, but show the same pattern: **bother and interference rise with age**, and disease-specific health-related QoL (HRQoL) is **worse with higher symptom scores** in population studies in the United States, Scotland, France and Japan.

Prostate Size

COMPARISON: METHODS OF MEASURING PROSTATE SIZE

Method	Performance
DRE	Poor inter-observer reliability ; significant error confirmed in a large screening trial; underestimates size, by 25% rising to 50% or more as glands enlarge

Method	Performance
TRUS (prolate ellipsoid formula)	Most widely accepted measure ; reasonable statistical performance, especially with a trained examiner
Three-dimensional TRUS	More accurate still; mostly unavailable in the office
MRI	Preferred with TRUS for research; expensive for population work; reads about 10% larger than TRUS
Computed tomography	Less accurate; reasonable precision for clinical purposes

- **In every cross-sectional study, TRUS volume rises slowly but steadily with age.** Differences in absolute values and slopes reflect the populations studied (Figure 149.15).
- **Overall pattern:** total prostate volume (TPV) rises from **about 25 mL in men in their 30s to 35–45 mL in their 70s**; transition zone volume (TZV) rises from **about 15 to 25 mL** over the same span.
- **The agreement between studies is striking** given their different measurement methods.

Measures of Obstruction

- **Invasive pressure–flow study is the best measure of subvesical obstruction.** Non-intubated free flow is at best an indirect measure of the probability of obstruction.
- **No large cross-sectional pressure–flow datasets exist**, because the test is invasive and costly, and none are likely.
- **Non-invasive penile cuff testing** approximates intravesical voiding pressure, gives results similar to invasive pressure–flow studies, and is increasingly used; a review of a very large number of tracings supports its utility.

MAXIMUM FLOW RATE (QMAX) AND PROBABILITY OF OBSTRUCTION

Qmax	Probability of obstruction
<10 mL/s	High
10–15 mL/s	Intermediate
>15 mL/s	Low

- **Three limitations reduce the usefulness of this scheme.**
 - **Qmax depends on voided volume.** Correction nomograms exist, but **none is universally accepted**.
 - **High diurnal variability.**
 - **High day-to-day within-subject variability.**
- **Qmax falls almost linearly with age.** In the Olmsted County Study the median fell from **20.3 mL/s at 40–44 years to 11.3 mL/s at 75–79 years**.
- **Combined thresholds** (at least moderate symptoms plus Qmax <15 mL/s) capture a steadily rising proportion of men from the fifth to the eighth decade, with similar results in Olmsted County and Spain.
- **Home flowmeters and smartphone applications** may enable future population-based flow studies.

Concept — *Histological BPH is nearly universal with age and identical across populations, whereas clinical BPH is whatever the chosen definition makes it. Each descriptive measure (symptoms, size, flow) rises with age, but none defines the disease on its own.*

Analytical Epidemiologic Studies

- **Without a clear disease definition, determinants may be sought for LUTS, prostate growth or BOO.**
- **Two determinants are established: functioning testes at puberty** (a permissive hormonal factor) and **age**, the most critical determinant of every aspect of the disease.

Religion

- **Jewish religion is associated with a higher prostatectomy rate** (about 2–2.5-fold) in several studies, but **not with a higher rate of BPH diagnosis**. The likely explanation is **greater health-care contact or readiness to be offered surgery**, not biology.

Socioeconomic Factors

- **Findings conflict**: higher BPH rates in higher income groups in one study, higher surgical rates in lower income groups in another. Better access may explain the first; readier acceptance of surgery the second.
- **Education and socioeconomic status do not affect IPSS responses**, but they **do affect expectations and perceived improvement**: higher-income patients need a **larger fall in score** to perceive the same improvement. Socioeconomic factors therefore act on **perception**, not growth or obstruction.
- **Olmsted County: retirement** was the one independent predictor of a high propensity to seek care after multivariable analysis. Age ≥ 65 , incomplete schooling and low income lost significance.
- **Boston Area Community Health (BACH) study**: **falling household income and difficulty paying for care** were associated with **more severe symptom clusters**.
- **Global burden, 1990–2019**: BPH burden rose in all socioeconomic groups, but the rise in incidence and years lived with disability was **far greater in low and middle socioeconomic quintiles**.
- **Rural versus urban residence** was **not associated** with recognised or unrecognised LUTS/BPH in the National Health and Nutrition Examination Survey (NHANES).
- **Symptom-based cluster analysis** identified remarkably **similar symptom clusters** in EpiLUTS and BACH, allowing analysis of comorbid associations.

TABLE 149.4 OVERVIEW OF THE EPI LUTS AND BACH STUDIES (From Kaplan et al., 2009)

Feature	EpiLUTS	BACH
Design	Cross-sectional, internet-based , population-representative survey	Population-based survey of randomly selected adults; longitudinal design
Setting and population	United States, United Kingdom, Sweden; men and women aged 40–99	Boston; men and women aged 30–79; Black, Hispanic and White groups
LUTS prevalence	Very common in both sexes when counted as occurring at least "sometimes"	AUA Symptom Index ≥ 8 in about one in five of both men and women ; rises with age; no difference by race or ethnicity
Bother	Lower for symptoms "sometimes" than "often". Leakage during sexual activity : infrequent but highly bothersome . Terminal dribble : frequent but less bothersome	Higher with LUTS than without, and higher in women than men
Treatment seeking	Low ; commonest in the voiding + storage + post-micturition subgroup, which also had the highest prescription use	Prescription use low even with AUA Symptom Index ≥ 8 , lower in women
Comorbidity	Commonest in the combined subgroup: arthritis, asthma, chronic anxiety, depression, diabetes (men only) , heart disease , irritable bowel syndrome, neurological disease, recurrent urinary tract infection (UTI), sleep disorders; childhood nocturnal enuresis linked to most LUTS subgroups	Heart disease, diabetes (men only) and depression

- **What the table shows**: two large studies with different designs agree that **LUTS are as common in women as in men**, are **under-treated**, and cluster with **cardiometabolic disease and depression**.

- **Trap:** frequency and bother do not run together. A rare symptom (leakage during sex) can be the most bothersome; a common one (terminal dribble) may bother little.

Alcohol and Liver Cirrhosis

- **Alcohol lowers plasma testosterone** (reduced production, increased clearance), which would predict less BPH.
- **Observed associations are inverse:** heavy drinking was associated with **lower rates of surgery for BPH**.
 - **Caveat:** poorer health in heavy drinkers may bias physicians against surgery. One study found no association with diagnosis or surgery.
- **Prostate Cancer Prevention Trial (PCPT):** an **inverse relationship** between alcohol intake and BPH diagnosis, that is, a modest protective effect.
- **Meta-analysis:** alcohol **reduces the likelihood of BPH but not of LUTS**. Higher intake (≥ 36 g/day) against none was associated with a **lower likelihood of BPH (odds ratio 0.65)**.
- **Cirrhosis:** most autopsy studies found **less BPH in cirrhosis**, but because most cirrhosis is alcohol-induced, the two effects **cannot be separated**.
- **Non-alcoholic fatty liver disease (NAFLD)**, part of the metabolic syndrome, is associated with **BPH in men and OAB in women**.

Sexual Activity and Vasectomy

- **The hypothesis that sexual activity increases fibromuscular stroma** is unsupported.
 - An apparent protective association of widowhood against prostatectomy was not replicated.
 - **Olmsted County:** ejaculatory frequency has **no effect on LUTS, Qmax or prostate volume**; the apparent protective effect is **confounded by age**.
 - The coincident decline of sexual activity and rise of BPH with age may reflect a **reverse relationship:** BPH impairing sexual function.
- **LUTS severity correlates strongly with erectile and ejaculatory dysfunction** (Figure 149.16).
 - Erectile function **falls** and ejaculatory disturbance **rises** with age, and **within every decade** also with worsening LUTS.
 - Correlation does not prove causation. A **shared mechanism, most obviously ischaemia** of genital and lower urinary tract organs, is plausible.
- **Vasectomy is not a risk factor for BPH**.
 - An initially raised risk ratio became **non-significant after 5 years** of follow-up.
 - No relationship with BPH or prostate size; the **Massachusetts Male Aging Study (MMAS)** found no increased risk of BPH diagnosis.

TABLE 149.5 LOGISTIC REGRESSION OF DECREASED SEXUAL ENJOYMENT AND ACTIVITY AS WELL AS PREDICTORS OF ERECTILE DYSFUNCTION, EJACULATORY DYSFUNCTION, AND PREMATURE EJACULATION (From Wein et al., 2009, EpiLUTS)

Predictor group	Decreased sexual enjoyment	Sexual activity	Erectile dysfunction (IIEF-EF <21)	Ejaculatory dysfunction	Premature ejaculation
Age	—	—	+	+	—
Voiding LUTS	Weak stream +, split stream +	Weak stream +	Weak stream +, split stream +	—	Intermittency +, terminal dribble +
Storage LUTS	Perceived frequency +, urgency with fear of leaking +	Perceived frequency +	Urgency with fear of leaking +	Urgency with fear of leaking +	—

Predictor group	Decreased sexual enjoyment	Sexual activity	Erectile dysfunction (IIEF-EF <21)	Ejaculatory dysfunction	Premature ejaculation
Leakage	Leak during sex +; stress leakage (laugh/cough) –	Leak during sex +; leak for no reason +	Leak during sex +	Leak during sex +	—
Emptying and pain	Incomplete emptying +, dysuria +, bladder pain +, pain during sex +	Incomplete emptying +, dysuria +, pain during sex +	Dysuria +	—	Incomplete emptying +, bladder pain +
Comorbidity	Prostate cancer +, bladder cancer +, prostatitis +	Heart disease +, prostate cancer +, bladder cancer +, prostatitis +	Hypertension +, diabetes +, depression +	Prostate cancer +, depression +	Prostatitis – (absence of prostatitis predicts it)
Model concordance index	0.86	0.86	0.77	0.73	0.62

- **What the table shows:** specific LUTS, **above all leakage during sex**, predict impairment in almost every sexual domain, independently of demographics.
- **Pattern:** erectile dysfunction tracks **cardiometabolic comorbidity** (hypertension, diabetes) and depression. Premature ejaculation is poorly predicted by LUTS (lowest concordance).
- **Trap:** age is a **positive** predictor of erectile and ejaculatory dysfunction but a **negative** predictor of premature ejaculation.

Hypertension

- **Sympathetic α -adrenergic pathways are common to hypertension and LUTS**, but both rise with age, so causality is **hard to prove**.
- **Most large studies found no association**; a small, flawed clinic study suggested one.
- **Autonomic hyperactivity is implicated in both LUTS and ED**, without conclusive clinical data.
- **EpiLUTS:** heart disease and hypertension were associated with **more severe symptom constellations**.

Depression

- **Depression is strongly associated with LUTS.** In NHANES, men with **moderate depression had the highest age-adjusted odds of clinical LUTS**. The association is confirmed by others, and **in women**.
- **Proposed link:** bladder inflammation and **central neuroinflammation**.
- **Post-finasteride syndrome:** depression, other mood disorders and even suicidal tendencies have been linked to finasteride use for LUTS/BPH.

Smoking

- **Nicotine raises testosterone and oestrogen**, which would predict more BPH; surgery is a poor end point because physicians avoid operating on heavy smokers.
- **Findings are inconsistent:**
 - No correlation with prostatectomy or diagnosis in some studies; an **inverse** association with prostatectomy and **smaller glands in smokers** in others.
 - **Olmsted County:** a **biphasic** relation (light-to-moderate smokers less likely to have moderate-to-severe LUTS; heavy smokers at least the same risk as never-smokers). Later: smoking associated with **lower flow and more symptoms**, not with prostate volume or PSA, and not with urinary retention.

- **EpiLUTS**: ex-smokers had higher, never-smokers lower, risk of severe symptom clusters; current smoking was not a factor.
- **BACH, a large meta-analysis and the REDUCE trial** (Reduction by Dutasteride of Prostate Cancer Events) found **no relationship**.
- **Bottom line**: any association is **weak and of limited clinical significance**.

Physical Activity, Diet, Obesity, Body Mass Index, and the Metabolic Syndrome

- **Diet**: no single dietary factor has a convincing major role. Isolated associations have been reported with beef, milk and low vegetable intake. A higher **dietary inflammatory index** was associated with **higher odds of clinical LUTS** in NHANES.
- **Metabolic syndrome**: obesity, glucose intolerance, dyslipidaemia and hypertension, raising cardiovascular risk and arising mainly from **physical inactivity and westernised diet**.
- **Biological plausibility**: **adipose tissue is the main site of aromatisation of testosterone to oestrogen**, and **leaner men have higher testosterone**.
- **Findings on body mass index (BMI) conflict**.
 - BMI was **inversely** associated with surgically treated or diagnosed BPH in two large cohorts.
 - Prostate weight and adenoma size **rose with obesity** (with rising oestradiol) in others, but **symptom severity did not**.
 - **Two biases**: DRE under-detects enlargement in the very obese, and obese men may be steered away from surgery.
- **Abdominal obesity is the relevant measure**. In the **Health Professionals Follow-up Study**, **waist circumference** predicted prostatectomy and frequent urinary symptoms **independently of BMI**; BMI, hip circumference and waist:hip ratio did not.
- **Hyperinsulinaemia hypothesis**: non-insulin-dependent diabetes, hypertension, height, obesity, **high insulin and low high-density lipoprotein (HDL) cholesterol** were risk factors for BPH in a LUTS cohort. The proposed mechanism is **insulin-driven sympathetic overactivity**.
- **Physical activity** is associated with a **lower** risk of BPH and LUTS in pooled analysis of observational studies, at light, moderate and vigorous levels alike, and in EpiLUTS.
- **But a Cochrane review of randomised trials** found **very low-quality evidence**; whether exercise actually improves LUTS scores remains **uncertain**.
- **Current position**: inactivity, obesity, BMI and other metabolic syndrome measures are **positively associated** with LUTS and BPH (including volume). The **benefit of increasing activity is less clear**.

TABLE 149.6 COHORT STUDIES OF MODIFIABLE RISK FACTOR ASSOCIATED WITH DECREASED OR INCREASED RISK OF LOWER URINARY TRACT SYMPTOMS AND/OR BENIGN PROSTATIC HYPERPLASIA (From Parsons, 2007) — rebuilt by factor

Factor	Direction	Outcome	Thresholds compared	Odds ratio range
Alcohol	Decreased risk	Clinical BPH, TURP, nocturia, LUTS	Moderate intake (about 30–50 g/day) or heavy (≥ 60 g/day) or ≥ 1 drink/day vs none or < 5 g/day	0.40–0.80
Physical activity	Decreased risk	Clinical BPH, LUTS	Walking ≥ 2 h/week; > 6 sessions/week; MMAS: ≥ 862 kcal/day vs ≤ 140 kcal/day (OR 0.50)	0.49–0.73
Diabetes	Increased risk	LUTS	Diabetes vs none	1.25–2.80
Fasting glucose	Increased risk	LUTS; prostate ≥ 40 mL	> 110 mg/dL vs ≤ 110 mg/dL	2.60–2.98

Factor	Direction	Outcome	Thresholds compared	Odds ratio range
Obesity	Increased risk	LUTS; prostate ≥ 40 mL	BMI >35 vs <25 kg/m² (OR 3.52 for prostate ≥ 40 mL); weight gain since age 25	1.79–3.52
Abdominal obesity	Increased risk	BPH surgery; LUTS	Waist >109 cm vs <89 cm; waist >102 cm vs <94 cm; higher waist:hip ratio	1.32–2.38
Hypertension	Increased risk	LUTS	Hypertension vs none	1.29–1.76

- **What the table shows:** the modifiable factors fall cleanly into two groups. **Alcohol and exercise protect; the components of the metabolic syndrome** (diabetes, hyperglycaemia, obesity, central adiposity, hypertension) **increase risk**.
- **The strongest single signals** are obesity with a **large prostate** and **central adiposity with surgery**.
- **Trap:** the protective alcohol effect is on BPH, **not on LUTS**, and is confounded by physicians' reluctance to operate on heavy drinkers.

Medications

- **Evidence is very limited.**
- **Cold remedies containing α -sympathomimetics worsen LUTS** by stimulating bladder outlet smooth muscle.
- **Olmsted County:** daily antidepressants, antihistamines or bronchodilators add 2–3 IPSS points; daily antidepressants also lower flow rate.
- **Statins:** no significant effect on BPH incidence overall in a large meta-analysis, but a possible **reduced risk over age 60**.

Concept — *Apart from age and functioning testes, every candidate risk factor is confounded, because surgery is a biased end point and LUTS are not specific to the prostate. The one coherent signal is metabolic: central adiposity, diabetes and inactivity worsen LUTS and BPH, and activity and moderate alcohol intake appear protective.*

Correlations Between Parameters

- **All parameters worsen with age:** symptoms, bother, interference, HRQoL, Qmax and volume.
- **Correlations between them, and with pressure–flow findings, are generally weak.**
- **The exception is the subjective measures:** symptom score, bother, disease-specific HRQoL and interference correlate strongly with one another (Figure 149.17, which shows worse age-adjusted bother, interference, general health and sexual satisfaction with higher IPSS in the United States, Scotland, France and Japan).

Why correlations are weak.

- **Weak numerical correlations can arise artefactually**, whether or not a physiological relationship exists.
- **Variability:** within-individual, day-to-day, measurement error and the natural history itself.
- **Restriction of range:** correlations emerge most easily across the full spectrum of values. **Entry thresholds in BPH trials truncate the range and suppress correlation.**
 - **Figure 149.18:** in unselected volunteers, symptom score and Qmax correlate clearly. Restricted to typical trial entrants (**score >10 , Qmax 5–15 mL/s**), the correlation **virtually disappears** and the regression line is flat.
- **Weak numerical correlation does not preclude a strong clinical relationship.**

TABLE 149.7 CORRELATION TABLE AMONG BASELINE PARAMETERS IN A BENIGN PROSTATIC HYPERPLASIA POPULATION WITH INTERNATIONAL PROSTATE SYMPTOM SCORE >12, PEAK FLOW RATE (QMAX) <15 ML/SEC, SERUM PROSTATE-SPECIFIC ANTIGEN BETWEEN 1.5 AND 10 NG/ML, AND TOTAL PROSTATE VOLUME >30 ML

Pair	Strength of correlation
TPV with TZV	Strong (r about 0.78)
PSA with TPV / TZV	Moderate (r about 0.35–0.38)
Age with PSA, Qmax, TPV, TZV	Weak (r about 0.07–0.15)
IPSS with Qmax	Very weak (r about –0.12)
IPSS with TPV or TZV	Essentially none
Qmax with TPV or TZV	Very weak (r about –0.05)

- **What the table shows:** in a tightly selected trial population, **symptoms, flow and volume are almost independent**. Only **volume–volume** and **PSA–volume** correlations survive.
- **Trap:** a trial population is the worst place to test a correlation, because entry criteria restrict the range.

Community-based studies.

- **Correlations are somewhat stronger in true community samples** without entry criteria.
 - **Olmsted County:** after age adjustment, **enlargement (>50 mL)** gave about **3.5-fold** odds of moderate-to-severe symptoms, and **Qmax <10 mL/s** about **2.4-fold**. Men with enlarged prostates were about **twice as likely** to report bother or interference.
 - **Netherlands:** the IPSS correlated weakly but significantly with TPV, Qmax and post-void residual.

Transition zone.

- **Early reports suggested the transition zone, and especially the transition zone index (TZV/TPV), correlated better** with symptoms, flow and even detrusor pressure at peak flow. These came from a small cohort.
- **Larger datasets did not confirm this.** In Olmsted County, TZV correlated **no better than TPV** with symptoms or flow, and the transition zone index correlated only weakly. A large treatment trial showed the same.
- **A relationship between obstruction and prostate volume is rarely found;** most authors deny one in clinic and trial series.

Serum PSA and prostate volume.

- **Strong log-linear relationship** in population and clinical studies, but individual variability is too great to predict volume accurately in one patient.
- **Age modifies it: older men have more volume per unit of PSA** (Figure 149.19).
- **Asian (Japanese) men** show the same relationship with **smaller volumes and lower PSA** throughout.
- **Benign PSA (BPSA)** relates more strongly to BPH than total PSA.

Summary.

- **Apart from age, correlations are modest in the community and weak in clinic and trial populations.**
- **PSA–volume correlation is moderate** and influenced by age, race and ethnicity.
- **Neither symptoms, flow rate nor volume reliably predicts the presence or degree of obstruction.**

Concept — *Symptoms, flow, volume and obstruction are four separate dimensions that age together but predict one another poorly. Selecting patients by threshold makes the correlations weaker still, so obstruction can only be diagnosed by measuring it.*

Natural History of Untreated Benign Prostatic Hyperplasia

- **Natural history is the prognosis of a disease over time:** the change in parameters of interest and the incidence of significant outcomes.
- **It is the baseline against which every treatment is judged.** The benefits and risks of intervention must be balanced against the risks of simply watching.
- **How much natural history must be known depends on the seriousness of the disease and the risk of treatment.** The analogy is abdominal aortic aneurysm, where prognosis by size governs the decision to accept operative risk.

Clinical Parameters and Outcomes of Interest

COMPARISON: DIRECT VERSUS INDIRECT OUTCOMES

Direct (biological) outcomes	Indirect (proxy) outcomes
Noticed by the patient: symptom change, infection, death	Not noticed by the patient, but may predict future direct outcomes
—	Example: change in pressure–flow parameters predicting retention

Outcome scales and how they are summarised

Scale	Example	Summary measure
Dichotomous	Retention or no retention	Probability or rate
Categorical	Grade of obstruction on pressure–flow study	Probability or rate
Continuous	Symptom score, Qmax	Mean or median, with standard deviation, standard error or confidence interval

TABLE 149.8 BENEFICIAL AND ADVERSE CHANGES IN PARAMETERS AND OUTCOMES — rebuilt

Parameter or outcome	Beneficial change	Harmful change	Direct or indirect	Scale
Probability of symptom change	Improvement	Worsening	Direct	Dichotomous
Magnitude of symptom, bother, QoL change	Improvement	Worsening	Direct	Continuous
Flow rate, residual urine, prostate size, pressure–flow parameters	Improvement or reduction	Worsening or increase	Indirect	Continuous
Urinary incontinence, ED, haematuria, detrusor failure, upper tract deterioration	—	Occurrence	Direct	Dichotomous or categorical
AUR, need for treatment or surgery, UTI, bladder stones, bladder diverticula, death	—	Occurrence	Direct	Dichotomous
Azotaemia or renal failure	—	Occurrence	Direct	Continuous

- **What the table is for:** it is the vocabulary of every BPH trial. **Symptoms and events are direct; flow, residual, size and urodynamics are proxies.**
- **Trap:** a trial that improves Qmax or volume has improved a proxy, not an outcome the patient notices.

Methods of Studying Natural History of Benign Prostatic Hyperplasia

COMPARISON: THREE DESIGNS FOR STUDYING NATURAL HISTORY

Design	Population	Main biases
Watchful waiting cohorts	Diagnosed, untreated patients	Ethical and feasibility doubts; contact with carers alters behaviour ; progressors leave for treatment, depleting the cohort; selection by entry thresholds → unilateral regression to the mean
Control arms of randomised trials (no treatment, placebo against drug, sham against device or surgery)	Diagnosed patients in trials	As above, plus knowledge of being in the less effective arm (lower threshold to cross over) or a sense of obligation to finish first; regression to the mean from entry criteria
Longitudinal population-based studies	Unselected, undiagnosed community men, less likely to progress	No regression-to-the-mean bias , no diagnostic label; the best vehicle , though costly; monitoring itself still perturbs natural history

Regression to the mean.

- **Mechanism:** entry thresholds select men with higher scores and lower flows. The same thresholds are **not applied at follow-up**, so values drift back toward the mean. This **falsely suggests improvement**.
- **Demonstration:** untreated volunteers completed the IPSS and a flow test twice within a month.
 - **Unselected:** individual values moved in both directions, but **the means did not change**.
 - **Selected by trial-type thresholds** (IPSS above 7, 10 or 15; Qmax below 15, 12 or 10 mL/s): the second test showed **significant "improvement"** in both score and flow.

TABLE 149.9 MEAN $\pm$ STANDARD DEVIATION AND RANGE FOR AMERICAN UROLOGICAL ASSOCIATION SYMPTOM INDEX FOR TEST NO. 1 AND TEST NO. 2, MEAN DIFFERENCE BETWEEN THE TWO TESTS, AND THE 95% CONFIDENCE INTERVAL FOR THE DIFFERENCE, FOR ALL SUBJECTS AND FOR SUBJECTS CENSORED BASED ON TEST NO. 1 >7 POINTS, >10 POINTS, AND >15 POINTS (From Sech et al., 1998)

Selection at test 1	Mean change at test 2 (points)	Significant?	Subjects left
All subjects	-0.4	No	All
Score >7	-1.0	Yes	Fewer
Score >10	-1.1	Yes	Fewer
Score >15	-1.4	Yes	Fewest

- **What the table shows:** the **stricter the entry threshold, the larger the apparent improvement with no treatment at all**.
- **Censoring also shrinks the analysable group**, which reduces statistical power.
- **Implication:** part of the "placebo effect" in every BPH trial is a **purely mathematical conditional regression to the mean**, present whenever a threshold is set on a numerical scale.

Watchful Waiting Studies

- **Early watchful waiting cohorts are of limited rigour:** poorly defined entry criteria, poor follow-up and compliance, inadequate instruments and incomplete accounting. None used a quantitative symptom scale.
- **Their pooled findings** (five studies, 3–6 years):
 - Flow **deteriorated in about two-thirds** and improved in about one-fifth.
 - **Residual urine** rose, fell or stayed unchanged in roughly equal thirds; mean residual **rose**.
 - The meta-analysed **probability of any symptom improvement was about 42%**.

The Veterans Affairs (VA) Cooperative Study — landmark randomised trial of TURP against watchful waiting.

- **Design:** men with **moderate symptoms** (Madsen-Iversen score 10–20 on a 0–27 scale) randomised to **transurethral resection of the prostate (TURP) or watchful waiting**.
- **Treatment failure** was defined as: death, recurrent infection, **residual urine >350 mL**, bladder calculus, incontinence, **symptom score ≥ 24** , or **doubling of serum creatinine**.
- **3-year result:** failure was about **halved by TURP** (relative risk **0.48**).
 - **About one-quarter of the watchful waiting arm had surgery** within 3 years, most electively rather than for protocol failure. They were mostly the **more bothered** at baseline.
 - About **40%** of highly bothered men improved without surgery, matching the natural history estimate.
 - Men who stayed on watchful waiting without failing **improved slightly** in almost all outcomes.
- **5-year update:** **failure 21% with watchful waiting versus 10% with TURP**; all outcomes favoured TURP. The 5-year Kaplan–Meier crossover estimate was **36%**, and **elective crossovers outnumbered failure-driven crossovers two to one**.
- **High baseline bother was the strongest predictor of elective crossover** (Figure 149.20).

TABLE 149.10 EXTENDED FOLLOW-UP TREATMENT OUTCOMES BY INTENTION TO TREAT IN THE VETERANS AFFAIRS COOPERATIVE STUDY COMPARING WATCHFUL WAITING WITH TRANSURETHRAL RESECTION OF THE PROSTATE (From Flanigan et al., 1998)

Outcome	TURP (n = 280)	Watchful waiting (n = 276)	Relative risk (TURP vs WW)	Significant?
Treatment failure	28	58	0.48	Yes
Urinary retention	1	9	0.11	Yes
High residual urine	5	19	0.26	Yes
High symptom score	1	13	0.08	Yes
Death	16	12	1.31	No
Azotaemia	3	2	1.48	No
Incontinence	4	8	0.49	No
Bladder stones	0	1	—	No
Withdrawal of consent	57	36	1.56	Yes — more after TURP
Prostate cancer	24	11	2.15	Yes — more after TURP

- **Definitions:** high symptom score ≥ 21 on two consecutive measurements or ≥ 24 once; azotaemia doubling of baseline creatinine or >3.0 mg/dL.
- **What the table shows:** TURP prevents the **symptomatic and emptying failures** (retention, residual, symptom score). It does **not change death, azotaemia or incontinence**, which were rare in both arms.
- **Trap:** the excess **prostate cancer after TURP** reflects detection in resected tissue, not causation.

Delayed surgery gives less benefit.

TABLE 149.11 CHANGES FROM BASELINE TO 60 MONTHS OF FOLLOW-UP FOR PATIENTS INITIALLY RANDOMIZED TO TRANSURETHRAL RESECTION OF THE PROSTATE (1), RANDOMIZED AND REMAINING ON WATCHFUL

WAITING (2), AND THOSE RANDOMIZED TO WATCHFUL WAITING BUT CROSSING OVER TO TRANSURETHRAL RESECTION OF THE PROSTATE AT SOME POINT DURING THEIR FOLLOW-UP (3) (From Flanigan et al., 1998)

Measure (change)	Initial TURP: high / low baseline	Crossover TURP: high / low baseline	Watchful waiting: high / low baseline
Symptom score	-11.0 / -7.7	-8.0 / -6.2	-5.8 / -2.2
Qmax (mL/s)	+4.6 / +8.7	+2.9 / +4.7	-2.2 / +2.2
Residual urine (mL)	-100 / -19	-81 / -24	-50 / -7
Bother score	-34.9 / -14.7	-42.3 / -7.2	-10.7 / +1.6

- **What the table shows:** men who crossed over to TURP after a failed period of watchful waiting **improved less** than men operated on at once. The expected **fall of 11 points** (high baseline symptoms) and **rise of 8.7 mL/s** (low baseline flow) became **8 points** and **4.7 mL/s**.
- **Interpretation:** some **irreversible damage may occur during watchful waiting** before crossover.
- **Trap:** "high" and "low" refer to the baseline level of **that** measure; the largest gains accrue to the most abnormal baseline.

Progression of mild LUTS.

- **A clinic cohort with mild LUTS (IPSS <8)** under watchful waiting showed a **steadily rising cumulative probability of progression** to moderate (IPSS 8–18) or severe (IPSS 19–35) symptoms over 4 years, reaching about **one in three**.

Placebo and Sham Control Groups in Randomized Trials

Placebo control groups.

- **Guideline meta-analysis of placebo arms** (Agency for Health Care Policy and Research, 1994), compared with the watchful waiting studies.

TABLE 149.12 COMPARISON OF OUTCOMES AFTER WATCHFUL WAITING AND PLACEBO TREATMENT (Modified from McConnell et al., 1994)

Outcome	Watchful waiting (long-term)	Placebo (mean 13 weeks)
Flow rate increase / no change / decrease	About 20% / 14% / 66%	About 36% / 41% / 23%
Mean flow change	Small	+0.6 mL/s
Residual urine decrease / unchanged / increase	About 35% / 28% / 37%	About 38% / 26% / 36%
Mean residual change	Rise	Slight fall
Symptom improvement / unchanged / worse	About 42% / 26% / 32%	About 42% / 54% / 5%

- **What the table shows:** about **40% of patients report some improvement** without active therapy, whether watching or on placebo, and the **fluctuation in Qmax and residual is small**. This is the **background** against which active treatment is judged.
- **Why the designs differ:**
 - Placebo arms are **short** (3 days to 52 weeks), **blinded**, and patients have at least an even chance of receiving active drug.
 - Some trials tacitly or openly promise free active treatment or **earlier surgery** afterwards (a phenomenon unique to UK trials).
 - Watchful waiting patients knowingly accepted a conservative approach that might fail.
- **Trap:** placebo does not beat watching. Despite the added inactive preparation, the **probability of improvement is the same (about 40%)** in watchful waiting cohorts, the VA trial and placebo arms.

Sham control arms of device trials.

- **Sham arms** exist for balloon dilation (against sham cystoscopy), hyperthermia and **transurethral microwave thermotherapy (TUMT)**.
- **Sham symptom improvement was modest**; active thermotherapy gave **about twice the sham improvement**.
- **Exception**: in the multicentre **hyperthermia** trial, active treatment was **no better than sham**.
- **Sham changes in Qmax were trivial** (small rises or falls); active arms improved substantially, again except hyperthermia.

Placebo/sham effect and baseline severity.

- **The placebo response rises with baseline symptom severity**. In a 12-month terazosin trial the placebo improvement increased stepwise across baseline severity strata, while active drug gave **about twice the improvement within each stratum**. The finasteride phase III trials showed the same.
- **Explanation**: greater expectation in more symptomatic men, or **regression to the mean**. The sham effect in device trials probably behaves alike.

Natural history and disease progression in long-term placebo arms.

- **When a placebo arm runs long enough, natural history emerges beneath the placebo response.**
- **Proscar Long-Term Efficacy and Safety Study (PLESS)** — landmark: men with **moderate symptoms and enlarged glands, finasteride 5 mg daily versus placebo for 4 years**.
 - After the initial placebo response, **mean symptom score and Qmax drifted back to baseline**. Changes after the initial response represent **natural history**.
 - **AUR, surgery and volume change are little affected by placebo**, so they are valid natural history measures.
 - Only a minority had MRI volumes, so **serum PSA was used as a proxy for volume**.

COMPARISON: PLESS PLACEBO ARM BY BASELINE PSA TERTILE (FIGURE 149.21)

PSA tertile	Symptoms over 4 years	Qmax over 4 years
Lowest (0–1.3 ng/mL)	Placebo response maintained	Response maintained
Middle (1.4–3.2 ng/mL)	Slow drift back to baseline	Gains negated
Highest (3.3–10 ng/mL)	Initial fall of 1.5 points, then rise of about 0.5 points per year to baseline	Gains lost by 2 years; net fall of 1.0 mL/s at 4 years

- **Volume tertiles (14–41, 42–57, 58–150 mL)** gave the same patterns.
- **Baseline PSA also predicts deterioration in bother, disease-specific QoL and some aspects of sexual function** after the initial placebo response (Figure 149.22).

Medical Therapy of Prostatic Symptoms (MTOPS) — landmark progression study.

- **Design**: placebo versus **doxazosin** versus **finasteride** versus **combination**, 4–5 years.
- **Composite progression end point**: **IPSS rise ≥ 4 points confirmed within 4 weeks, AUR, socially unacceptable incontinence, recurrent UTI, or renal insufficiency due to BPH**. A 4-point rise was chosen as a **clinically meaningful** worsening.
- **Paradox**: the placebo group improved overall (**IPSS -4.9**), yet **about 14% had a confirmed ≥ 4 -point worsening**.

TABLE 149.13 CLINICAL PROGRESSION OF BENIGN PROSTATIC HYPERPLASIA IN PLACEBO-TREATED PATIENTS IN THE MEDICAL THERAPY OF PROSTATIC SYMPTOMS TRIAL (From McConnell et al., 2003)

Event	Rate per 100 patient-years	Cumulative incidence at 4 years
Clinical progression overall	4.5	17%
IPSS rise ≥ 4 points	3.6	14%
Acute urinary retention	0.6	2%
Socially unacceptable incontinence	0.3	<1%
Urinary tract infection	0.1	<1%
Renal insufficiency due to BPH	0	0
Invasive therapy for BPH	1.3	5%

- **What the table shows:** symptom progression is by far the commonest form of progression. AUR and invasive therapy are uncommon, and renal insufficiency from BPH did not occur at all.
- **Trap:** progression in MTOPS is **mostly symptomatic**, not retention or surgery.

REDUCE placebo arm — landmark.

- **REDUCE was a cancer prevention trial**, so its placebo arm is **free of a LUTS placebo effect**.
- **Prostate size predicts incident LUTS:** 40–80 mL carried a **67% higher risk (hazard ratio 1.67)** than ≤ 40 mL.
- **Qmax predicts incident LUTS:** against ≥ 15 mL/s, 10–14.9 mL/s (hazard ratio 1.39) and < 10 mL/s (hazard ratio 1.67) carried higher risk.

Individual risk prediction.

- **Pooled placebo and active arms of dutasteride and dutasteride–tamsulosin trials** were used to build **nomograms and a web-based tool** predicting IPSS change, **retention and surgery** on placebo or active treatment from individual baseline values. It has been updated to predict **storage, voiding, QoL and nocturia** separately (Figure 149.23).

COMPARISON: THE LANDMARK TRIALS THAT DEFINE BPH NATURAL HISTORY

Trial	Design	Population	Natural history contribution
VA Cooperative Study	TURP vs watchful waiting	Moderate symptoms	Watchful waiting failure about twice TURP; delayed TURP gives less benefit ; bother drives crossover
PLESS	Finasteride vs placebo, 4 years	Moderate symptoms, enlarged gland	PSA and volume predict symptom, flow, AUR and surgery trajectories
MTOPS	Placebo vs doxazosin vs finasteride vs combination, 4–5 years	Clinical BPH	Progression 4.5 per 100 patient-years , mostly symptomatic; no renal insufficiency
REDUCE	Dutasteride vs placebo (cancer prevention)	Men at risk of prostate cancer	Size > 40 mL and Qmax < 15 mL/s predict incident LUTS , without placebo bias

Relationship between placebo/sham effect and perception of improvement.

- **A mean fall of about 3 AUA Symptom Index points corresponds to "slight" improvement**, but the fall needed depends strongly on **baseline severity**.

SYMPTOM SCORE CHANGE NEEDED TO PERCEIVE IMPROVEMENT (LOWER VS HIGHER BASELINE SCORE)

Perceived change	Lower baseline	Higher baseline
Marked improvement	–7.4	–15.3
Moderate improvement	–4.0	–8.7

Perceived change	Lower baseline	Higher baseline
Slight improvement	-1.9	-6.1
No improvement	-0.2	-2.0
Worsening	+3.3	+1.2

- **Trap:** the more symptomatic the man, the **larger the fall he needs to notice benefit**. A fixed point change does not mean the same thing to every patient.

Longitudinal population-based studies.

- **Community men aged 60 and over:** surgery about 3% per year; symptoms common and incident each year; but **spontaneous remission** also occurs, and about **one in five with severe symptoms were asymptomatic** a year later.

TABLE 149.14 CHANGES IN STATUS OF SYMPTOM SEVERITY FROM ONE SURVEY TO THE NEXT (Modified from Diokno et al., 1992)

Severity this year	Most common state next year	Pattern
None	None (large majority)	Stable
Mild	Mild (about half)	Moves in both directions; many become asymptomatic
Moderate	Spread across all categories	Highly unstable
Severe	Severe (about half)	About one in five become asymptomatic

- **What the table shows: fluctuation, spontaneous remission and regression to the mean.** Men who underwent prostatectomy were excluded from the next survey.
- **Forth Valley, Scotland** (BPH defined as **TRUS volume >20 mL, Qmax <15 mL/s** and a symptom threshold):
 - Prevalence rose steeply from the 40s to the 60s, and about half of affected men reported interference with daily activities.
 - Over 3 years, **moderate symptoms, eight of twelve individual symptoms, bother and interference all increased** (Figure 149.24).
 - **LUTS and BPH are progressive, but the slope is very flat.**

The Olmsted County Study of Urinary Symptoms and Health Status Among Men — the most informative natural history study.

- **Symptoms:**
 - A substantial minority of men with none-to-mild symptoms developed moderate-to-severe symptoms within 42 months, but **many reverted**.
 - The mean **IPSS rose by about 0.2 points per year** at 42 months, and by **0.34 points per year** at 92 months, with about **one-third showing a ≥3-point rise**.
 - **The steepest rise is in the 60s** (about 0.4–0.6 points per year), less in the 40s and the 70s.
- **Flow:** median **Qmax falls about 2.1% per year**.
 - Decline is faster with lower baseline flow and with older age, larger volume and worse symptoms.
 - **After mutual adjustment**, rapid decline (≥4.5% per year) is predicted only by **age ≥70** and **baseline Qmax <10 mL/s**, not by volume or symptoms.
- **Volume:** growth about **0.6 mL per year (6 mL per decade)** at ages 40–79, following an **exponential** pattern (**0.4 mL per year at 40–59, 1.2 mL per year at 60–79**).
 - An updated median of **about 1.9% per year**, independent of age and symptoms.
 - **Higher baseline PSA and larger volume predict faster growth.**

BACH 5-year follow-up.

- **Symptom progression (AUA Symptom Index rise >3)** occurred in roughly a quarter to a third; **regression (fall ≥3)** in a third or more, most often in **women and Hispanic participants**.
- **Age and higher BMI predicted progression, but not regression.**

TABLE 149.15 CHANGE IN LOWER URINARY TRACT SYMPTOMS SEVERITY CLASSIFICATION: AGE ADJUSTED PERCENTAGE OF PATIENTS WITH LOWER URINARY TRACT SYMPTOMS SEVERITY AT FOLLOW-UP BY LOWER URINARY TRACT SYMPTOMS SEVERITY AT BASELINE (From Maserejian et al., 2013)

Baseline severity	Men at follow-up	Women at follow-up
None	Most move to mild ; about a quarter stay none	Most move to mild; more stay none than in men
Mild (the commonest baseline state)	About three-quarters stay mild ; about one in ten each improve or worsen	Most stay mild; slightly more worsen than in men
Moderate	Roughly half stay moderate; most of the rest improve to mild	Similar to men; a minority become severe
Severe (rare)	Most improve	Most improve, largely to moderate

- **What the table shows:** in the community, **LUTS move in both directions**. Progression is real but **regression is as common**, and the pattern is the same in both sexes.

Concept — *Untreated BPH progresses slowly and unevenly: symptoms fluctuate and often remit, while volume and flow change steadily. Selection and regression to the mean inflate apparent improvement in every trial, so PSA, volume and flow, rather than symptoms, are the best markers of which men will progress.*

Complications of Benign Prostatic Hyperplasia

- **Most complications are rare**, and knowledge comes mainly from **case series of men presenting with them** rather than from cohorts observed for their development.

Mortality

- **BPH mortality has collapsed.** In 1950–1954, most reporting countries had rates **>10 per 100,000**; by 1985–1989 **only one of 61** did. Applying 1950 rates to 1990 implies **many thousands of deaths avoided** in the United States alone.
- **LUTS type predicts mortality in a long-term cohort:** **daytime frequency and nocturia** (especially when bothersome) were associated with **increased mortality**; **voiding LUTS with decreased mortality**.

Bladder Stones

- **Autopsy:** bladder stones are **about eight times commoner with histological BPH**, with **no increase in ureteric or renal stones**.
- **Clinically the risk is small:** one stone in the VA watchful waiting arm over 3 years.
- **Screen only when clinically indicated**, for example **haematuria or stuttering of the stream**.

Bladder Decompensation

- **The classic sequence:** normal mucosa → **trabeculation** → **cellules** → **diverticula** → **detrusor failure**.
- **When it starts, whether it is due to obstruction, and when intervention prevents it are all unknown.**
- **Trabeculated, obstructed bladders show dense connective tissue**, as in experimental obstruction; but **bladder fibrosis also occurs with age in both sexes**.
- **Is there a window for cure?** There is **no direct evidence** that delay causes irreversible loss. The indirect evidence is the **lesser benefit of delayed TURP in the VA trial**.
- **High-quality data are lacking;** randomised comparisons of **early versus deferred** intervention are needed.

Urinary Incontinence

- **Incontinence is among the most feared complications of BPH surgery.**
- **BPH causes it in two ways:**
 - **Overflow incontinence** from bladder overdistension.
 - **Urge incontinence** from **detrusor instability**, which affects **up to half or more of obstructed men**.
- **It is also age-related**, and **commoner in women** than men over 50 in the community.
- **Rates in BPH studies are low:** equal in both arms of the VA trial; **0.3 per 100 patient-years** (socially unacceptable incontinence) in MTOPS.

Urinary Tract Infections

- **UTI was the main surgical indication in about one-eighth** of older surgical series.
- **The link between residual urine and UTI is intuitive but unproven.**
- **Best data: MTOPS placebo arm, 0.1 UTI per 100 patient-years.**

Upper Urinary Tract Deterioration and Azotemia

- **Older series:** a mean of about **one in seven** men presenting for TURP had renal failure (range very wide), and about **one in thirteen** of those imaged had **hydronephrosis**, one-third of whom had renal insufficiency.
- **Renal failure raises TURP risk: higher complication rate and mortality up to six times higher.**
- **"Silent prostatism" (silent obstruction):** asymptomatic men who develop renal failure from BOO. **Rare but important.**
- **Modern cohorts:**
 - VA trial: azotaemia (**creatinine doubling**) in very few men in either arm.
 - **MTOPS: not one case of renal insufficiency due to BPH** in more than 3000 men over 4 years.
 - **Caveat:** trial screening may have excluded higher-risk men.
- **No cohort or population study has reported renal failure clearly attributable to BPH**, although self-reported episodes occur in cross-sectional surveys.

Hematuria

- **Gross haematuria and clots can arise from BPH alone.**
- **Men prone to it have higher prostatic microvessel density.**
- **Finasteride is a reasonable first-line therapy**, apparently by modulating **VEGF** expression.
- **No precise incidence data exist;** management depends on circumstances.

Acute Urinary Retention

- **AUR is one of the most significant outcomes of BPH.**
 - It was historically an **immediate indication for surgery** (about a quarter to a third of TURP indications in older series), and most men who fail catheter removal still have surgery.
 - It is painful, costly and time-consuming: inability to void → emergency attendance → catheter → follow-up → trial of removal → recovery or surgery.
 - **Older literature:** recurrence in **56–64% within a week** of a first episode, and **76–83% with diagnosed BPH**.
- **Aetiology is poorly understood:** obstructive, myogenic and neurogenic mechanisms.
 - **Proposed precipitants:** prostatic infection, bladder overdistension, excessive fluid, alcohol, sexual activity, debility and bed rest.

- **Prostatic infarction** was proposed as a cause, but later studies found **similar infarction rates with and without AUR**.

COMPARISON: SPONTANEOUS VERSUS PRECIPITATED AUR

	Spontaneous AUR	Precipitated AUR
Definition	No trigger identified	Follows a trigger: non-prostatic surgery, catheterisation, anaesthesia, sympathomimetics, anticholinergics, antihistamines
Later spontaneous AUR	15%	9%
Later surgery	75%	26%

- **The distinction is prognostic but inconsistently applied** in the literature.
- **Trap:** precipitated AUR does **not** carry the same surgical prognosis. Only about a quarter need surgery.

Descriptive epidemiology.

- **Older estimates were wildly variable** (about 4 to 130 per 1000 person-years; 10-year cumulative incidence 4–73%).

TABLE 149.16 DESCRIPTIVE STUDIES ON THE INCIDENCE OF ACUTE URINARY RETENTION — rebuilt by setting

Setting	Study	Incidence (per 1000 person-years)
Community, unselected	Physicians' Health Study	4.5 in text; 5.2 crude, in Table 149.17 footnote
Diagnosed, untreated	Olmsted County Study (ages 40–79)	6.8
	VA Cooperative Study, watchful waiting arm	9.6
	Prostatectomy candidates electing observation	25
Placebo arms of BPH trials	PLESS (4 years)	18 (1.8 per 100 patient-years)
	Pooled 2-year trials	13.5
	MTOPS	6 (0.6 per 100 patient-years; 2% cumulative)
Older watchful waiting series	Three small series	3.7–130
Self-reported history	Spanish cross-sectional survey	— (5.1% ever)

- **What the table shows:** AUR incidence rises with how selected the population is: **lowest in unselected community men, higher in diagnosed men, highest in men with large glands and moderate symptoms**.
- **Constant hazard:** in the observed cohorts and placebo arms, AUR occurred at a **steady rate** over follow-up.
- **Two values for one cohort:** the Physicians' Health Study rate is **4.5** in the text and **5.2** in the table footnote. Both derive from the same 82 episodes in 15,851 person-years; 5.2 is the **crude** rate, so the 4.5 is presumably an adjusted rate.

Analytical epidemiology.

- **Age** is perhaps the most important risk factor: incidence rises **almost linearly from 40 to over 80** in European and American studies (Figure 149.25).
- **Recurrence:** in a very large database, about a third of men with AUR had **three or more further episodes**. Risk of recurrence rises with **older age, White race, diabetes, neurological disease and low income**.

- **Symptoms:** greater severity means greater AUR risk (Table 149.17).
 - **All seven IPSS items individually predict AUR.**
 - **Best independent predictors:** incomplete emptying, voiding again within 2 hours, and weak stream.
 - **Medications with adrenergic or anticholinergic effects** also predict AUR.

TABLE 149.17 INCIDENCE OF ACUTE URINARY RETENTION BY PATIENT AGE AND AMERICAN UROLOGICAL ASSOCIATION SYMPTOM INDEX SCORE IN TWO POPULATION-BASED STUDIES (From Kaplan et al., 2008) — rebuilt

Age	Physicians' Health Study: AUASI ≤7	Physicians' Health Study: AUASI >7	Olmsted County: AUASI ≤7	Olmsted County: AUASI >7
40s	0.4 (45–49)	3.3 (45–49)	2.6	3.0
50s	1.2	10.0	1.7	7.4
60s	3.6	14.1	5.4	12.9
70s	7.9 (70–83)	11.3 (70–83)	9.3	34.7

- **Units:** AUR events per 1000 man-years.
- **What the table shows: age and symptoms multiply.** The highest rate is in men in their 70s with more than mild symptoms (Olmsted).
- **Trap:** even men with **mild symptoms** have a rising AUR rate with age.

Urodynamic parameters (Olmsted County, Figure 149.26).

RELATIVE RISK OF AUR (BASELINE 1.0)

Risk factor	Relative risk
Moderate-to-severe symptoms (IPSS ≥8)	3.2
Qmax <12 mL/s	3.9
Prostate volume >30 mL (TRUS)	3.0
Age 60–69 + more than mild symptoms + Qmax <12 mL/s	10.3
Age ≥70 (except mild symptoms with Qmax >12 mL/s)	12.9–14.8

- **Reference group:** men aged 40–49 with mild symptoms and Qmax >12 mL/s.

Prostate volume and serum PSA.

- **In diagnosed BPH populations, volume, PSA and symptoms all predict AUR.**
- **PLESS:** over 4 years, AUR incidence rose with symptom severity within each PSA stratum, and was higher at PSA >1.4 ng/mL than below.
- **Pooled 2-year trials:** AUR was **several-fold higher with PSA >1.4 ng/mL** and **about threefold higher with volume >40 mL.**
- **Risk rises almost linearly with increasing PSA thresholds** (Figure 149.27), for **both spontaneous and precipitated AUR**, and with **PSA or volume tertile** (Figure 149.28).
- **MTOPS confirmed rising AUR risk with PSA and volume** (Figure 149.29), as did the 2-year dutasteride trials.
- **Among more than 100 candidate predictors**, a combination (PSA, voiding more often than every 2 hours, symptom problem index, Qmax, hesitancy) was **only slightly better than PSA alone.**

Surgery for Benign Prostatic Hyperplasia

COMPARISON: AUR VERSUS SURGERY AS END POINTS

AUR	Surgery
An outcome that mandates management	One management option
Like a bone fracture : the physician cannot make it more or less likely, and cannot undo it	Influenced by the consultation : interaction style, quoted probabilities, local practice
A hard end point	A softer end point, varying by provider, region, health plan and era
Accounts for 5% to over 30% of surgical indications, depending on practice	Most surgery is for symptoms , not AUR

- **Analogy**: myocardial infarction and coronary artery bypass grafting. Not every infarct leads to bypass, and bypass is often done for other reasons, so bypass rates vary geographically. The same **variation** prompted the Agency for Health Care Policy and Research to develop BPH guidelines.
- **TURP and laser ablation remain the commonest and best-studied operations.**

Medicare trends.

- **TURP made up over half of major urological surgery in 1962 and just over a third by 1986.**
- **Medicare TURPs peaked in 1987 and fell by about a third by 1993**, yet TURP remained **second only to cataract surgery** among Medicare's most costly procedures.
- **TURP rates rise with age** in men over 65.
- **30-day mortality after TURP for BPH fell from 1.20% (1984) to 0.77% (1990).**
- **Age-adjusted TURP rates fell by a further 40–50% during 1991–1997.**
- **Limitation**: Medicare data cover only men over 65.

Risk of surgery in cohorts.

- **Some older series projected unrealistically high surgery rates.**
- **Community men ≥ 60** : surgery **about 3% per year**; frequency, hesitancy, straining and interrupted stream increased the risk.
- **Baltimore Longitudinal Study of Aging (BLSA)** — the first high-quality study: more than 1000 men followed for 30 years.
 - **Independent predictors**: age, incomplete emptying, change in size and force of stream, and enlargement on DRE.
 - **Cumulative surgery by number of risk factors**: none 3%, one 9%, two 16%, three 37%.
- **VA Normative Aging Study**: nocturia and hesitancy predicted surgery.
- **Kaiser Permanente**: age and five LUTS (dysuria, incontinence, difficulty starting, nocturia, slow stream) predicted surgery.
- **VA Cooperative Study**: about a quarter of the watchful waiting arm crossed over within 3 years; **high baseline bother** predicted surgery.
- **Diagnosed men electing observation**: 4-year probability of surgery rose steeply from mild to moderate to severe baseline symptoms.
- **Olmsted County Study**:
 - Treatment incidence **16 per 1000 person-years**, rising from **3.3 in the 40s to >30 per 1000 person-years at ≥ 70 .**
 - **Moderate-to-severe symptoms, $Q_{max} < 12$ mL/s, volume > 30 mL or PSA ≥ 1.4 ng/mL** each gave **about four times** the risk of treatment.

- **After mutual adjustment:** symptoms (**hazard ratio 5.3**), low flow (**2.7**) and enlargement (**2.3**) each predicted treatment independently.
- **Nearly one man in four was treated in his eighth decade.**
- The same factors predict treatment and AUR, with risks of similar size.
- **PLESS placebo arm: 10% had surgery over 4 years (2.5% per year), at a constant hazard.**
 - By **PSA tertile**, surgery rose from **6.2% to 14.6%**; by **volume tertile**, from **6.7% to 14.0%** (Figure 149.30).
- **MTOPS placebo arm:** invasive therapy rose **almost linearly** with baseline PSA and volume (Figure 149.31).
- **The web-based prediction tool** estimates retention and surgery risk from age, IPSS, Qmax, residual, volume and PSA.

Concept — *The serious complications of BPH (renal failure, stones, infection, death) have become rare. What remains common is AUR and surgery, and both are predicted by the same factors: age, symptoms, low flow and, above all, a large gland as reflected by PSA. AUR is the hard end point; surgery is soft and varies with practice.*

KEY POINTS — EPIDEMIOLOGY AND NATURAL HISTORY

- ****Autopsy prevalence of BPH rises linearly from age 40**** and is ****remarkably similar across cultures and ethnicities****. - ****LUTS severity, frequency, bother and QoL impairment rise with age****, fundamentally alike in all societies studied. - ****Analytical studies have failed to identify significant risk factors**** for BPH and LUTS. - ****Natural history varies widely between individuals****, measured chiefly by ****prostate volume growth****. - ****Variability also affects**** flow, residual urine, complications, ****retention and the need for surgery****. - ****Certain baseline factors allow assessment of risk**** of volume growth, progression and complications.